

Q1: An electron and proton are separated by a large distance. The electron starts approaching the proton with energy 3 eV. The proton captures the electrons and forms a hydrogen atom in second excited state. The resulting photon is incident on a photosensitive metal of threshold wavelength $4000 \text{ \AA}$. What is the maximum kinetic energy of the emitted photoelectron?

- (A) 7.61 eV
- (B) 1.41 eV
- (C) 3.3 eV
- (D) No photoelectron would be emitted

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Solution:

$$\phi_0 = 3.1 \text{ eV}$$

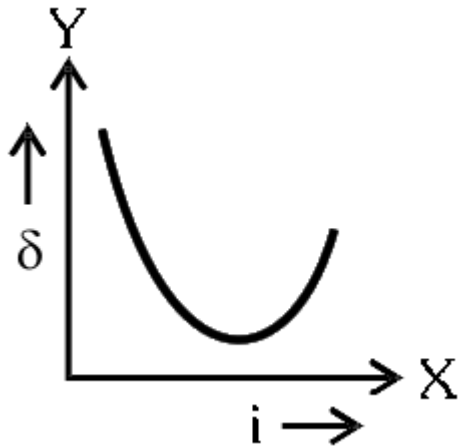
$$E_1 = \frac{-13.6}{9} \text{ eV}$$

$$\therefore E_{\text{photon}} = \left(3.0 + \frac{13.6}{9} \right) \text{ eV} = 4.511$$

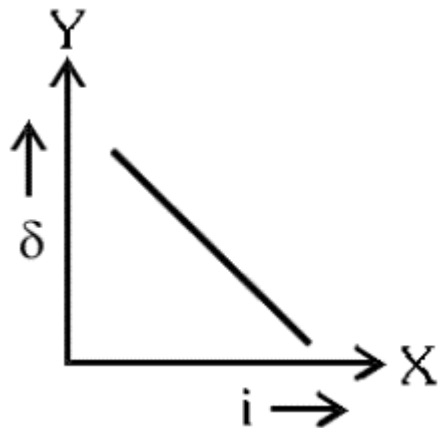
$$\therefore \text{KE of photoelectron} = 4.511 - 3.1 = 1.41 \text{ eV}$$

Q2: The expected graphical representation of the variation of angle of deviation ' δ ' with angle of incidence ' i ' in a prism is:

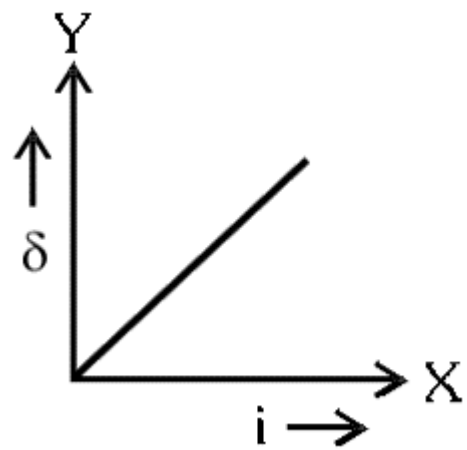
(A)



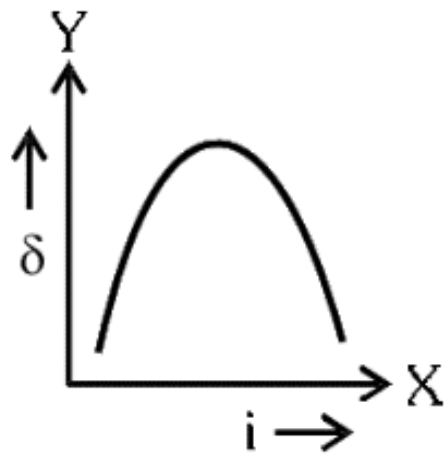
(B)



(C)

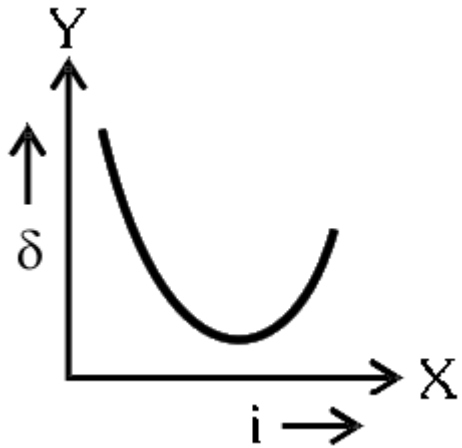


(D)

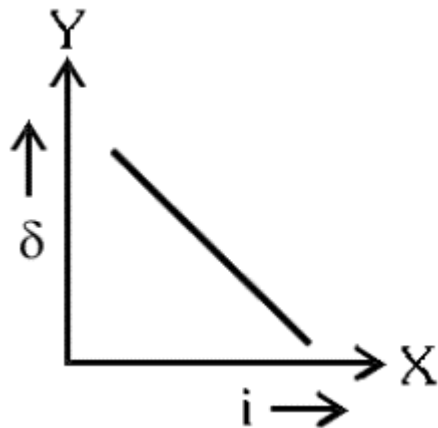


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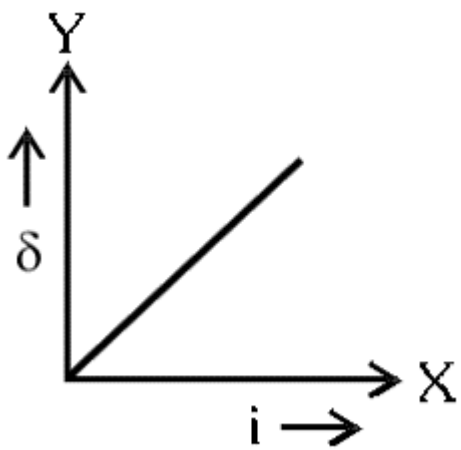
(A)



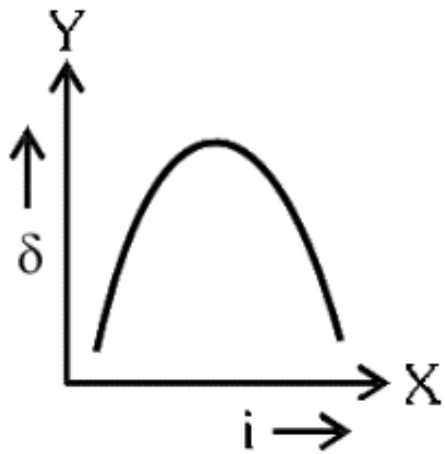
(B)



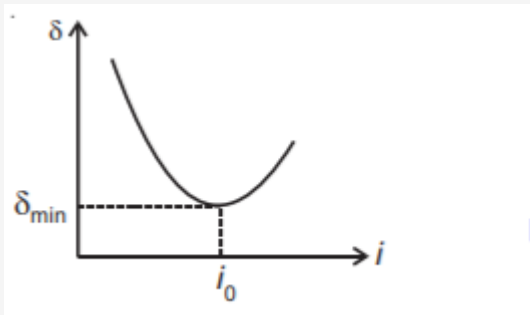
(C)



(D)



Solution:



Q3: A raindrop with radius $R = 0.2 \text{ mm}$ falls from a cloud at a height $h = 2000 \text{ m}$ above the ground. Assume that the drop is spherical throughout its fall and the force of buoyance may be neglected, then the terminal speed attained by the raindrop is:

[Density of water $\rho_w = 1000 \text{ kg m}^{-3}$ and Density of air $\rho_a = 1.2 \text{ kg m}^{-3}$, $g = 10 \text{ m/s}^2$
Coefficient of viscosity of air $= 1.8 \times 10^{-5} \text{ Nsm}^{-2}$]

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- (B) 43.56 ms^{-1}
- (C) 4.94 ms^{-1}
- (D) 14.4 ms^{-1}

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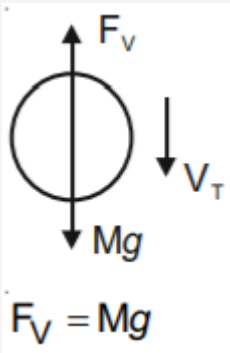
(A) 250.6 ms^{-1}

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(C) 4.94 ms^{-1}

(D) 14.4 ms^{-1}

Solution:



$$6\pi\eta R V_T = \left(\frac{4}{3}\pi R^3\right) \rho_w \times g$$

$$V_T = \frac{2R^2 \rho_w g}{9 \times \eta} = \frac{2 \times (2 \times 10^{-4})^2 \times 1000 \times 10}{9 \times 1.8 \times 10^{-5}}$$

$$\simeq 4.94 \text{ m/s}$$

Q4: One mole of an ideal gas is taken through an adiabatic process where the temperature rises from 27°C to 37°C . If the ideal gas is composed of polyatomic molecule that has 4 vibrational modes, which of the following is true? [$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$]

- (A) work done by the gas is close to 332 J
- (B) work done on the gas is close to 582 J
- (C) work done by the gas is close to 582 J
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Solution:

$$\Delta T = 37 - 27 = 10^{\circ}\text{C}$$

$$f = 3 + 3 + 4 \times 2 = 14$$

$$\therefore \gamma = 1 + \frac{2}{f} = 1 + \frac{2}{14} = \frac{8}{7}$$

$$\therefore W = \frac{-nR\Delta T}{(\gamma-1)} = \frac{-1 \times 8.314 \times 10}{\left(\frac{1}{7}\right)} = -582 \text{ J}$$

Q5: An object of mass 0.5 kg is executing simple harmonic motion. Its amplitude is 5 cm and time period (T) is 0.2 s. What will be the potential energy of the object at an instant $t = \frac{T}{4}$ s starting from mean position. Assume that the initial phase of the oscillation is zero.

- (A) 0.62 J
- (B) $6.2 \times 10^{-3} \text{ J}$
- (C) $1.2 \times 10^3 \text{ J}$
- (D) $6.2 \times 10^3 \text{ J}$

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Solution:

$$m = \frac{1}{2} \text{ kg} \quad A = 5 \text{ cm}, \quad T = 0.2 \text{ s}$$

$$x = A \sin \omega t$$

$$\text{P.E} = \frac{1}{2} m \omega^2 A^2 \sin^2 \omega t$$

$$= \frac{1}{2} m \omega^2 A^2 \sin^2 \frac{2\pi}{T} \times \frac{T}{4}$$

$$= \frac{1}{2} m \omega^2 A^2$$

$$= \frac{1}{2} \times \frac{1}{2} \times \frac{4\pi^2}{(0.2)^2} \times \left(\frac{5}{100} \right)^2$$

$$= \frac{\pi^2}{16} \approx 0.62 \text{ J}$$

Q6: Match List I and List II.

List – I List – II

(a) Capacitance, C

(i) $M^1 L^1 T^{-3} A^{-1}$

(b) Permittivity of free space, ϵ_0

(ii) $M^{-1} L^{-3} T^4 A^2$

(c) Permittivity of free space, μ_0

(iii) $M^{-1} L^{-2} T^4 A^2$

(d) Electric field, E

(iv) $M^1 L^1 T^{-2} A^{-2}$

Choose the correct answer from the options given below

(A) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (i)

(B) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (i)

(C) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (i)

(D) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (i)

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Solution:

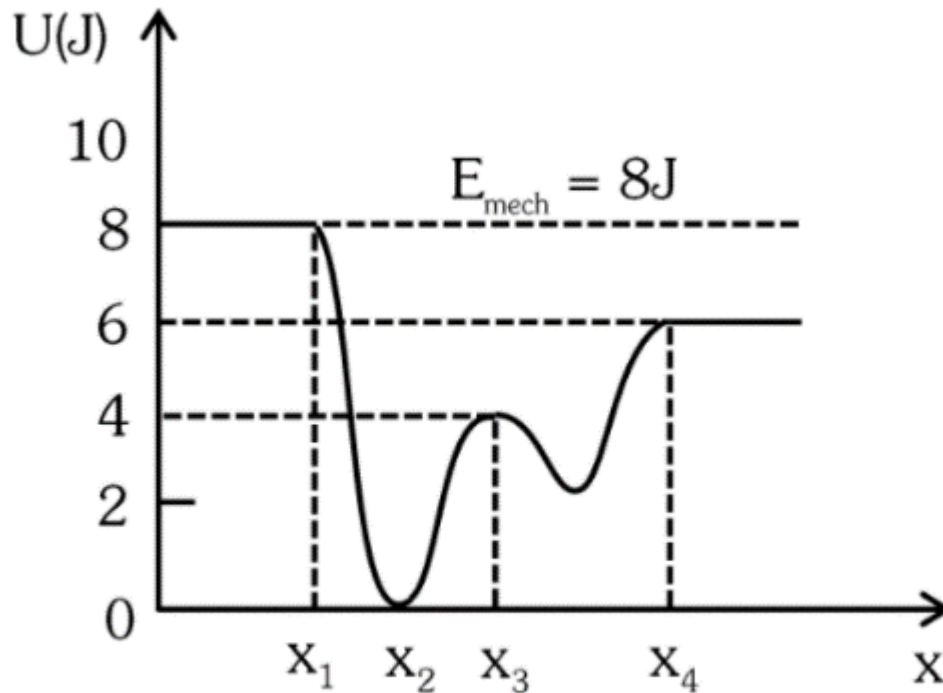
$$(a) U = \frac{Q^2}{2C} \Rightarrow [C] = \frac{I^2 t^2}{U} = \frac{A^2 T^2}{[ML^2 T^{-2}]} = [M^{-1} L^{-2} T^4 A^2]$$

$$(b) [\varepsilon_0] \rightarrow [M^{-1} L^{-3} T^4 A^2]$$

$$(c) [\mu_0] \rightarrow [MLT^{-2} A^{-2}]$$

$$(d) W = qE \times d \Rightarrow E = \frac{[ML^2 T^{-2}]}{[AT][L]}$$

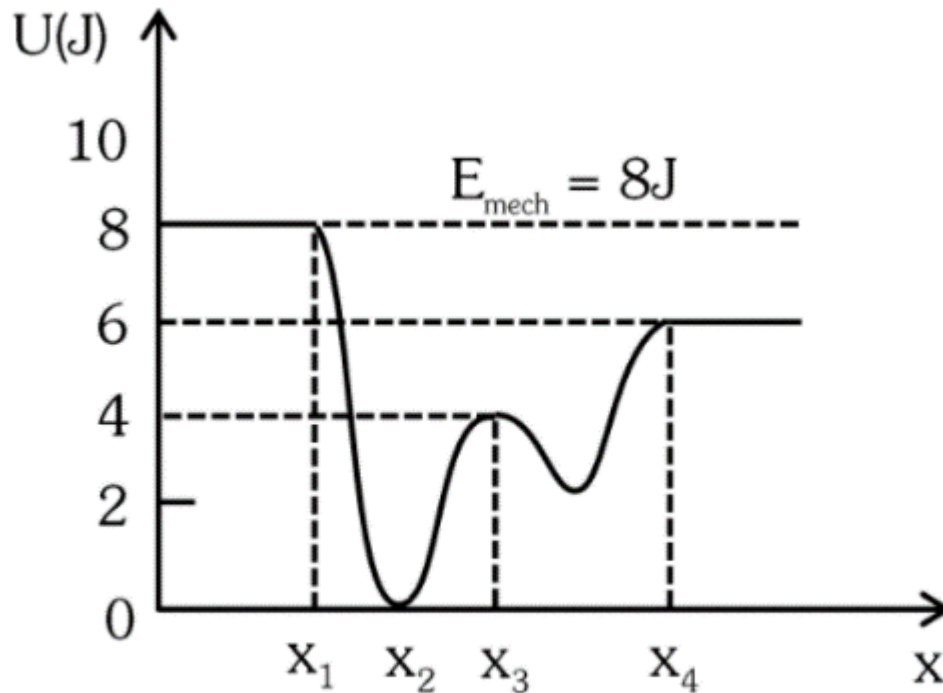
Q7: Given below is the plot of a potential energy function $U(x)$ for a system, in which a particle is in one dimensional motion, while a conservative force $F(x)$ acts on it. Suppose that $E_{mech} = 8 \text{ J}$, the incorrect statement for this system is:



[where K.E. = kinetic energy]

- (A) at $x > x_4$, K.E. is constant throughout the region.
- (B) at $x < x_1$, K.E. is smallest and the particle is moving at the slowest speed.
- (C) at $x = x_2$, K.E. is greatest and the particle is moving at the fastest speed.
- (D) at $x = x_3$, $K.E. = 4 J$.

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Solution:

At $x < x_1$, $KE = 0$
 $\Rightarrow$ Particle is at rest

Q8: A $100\ \Omega$ resistance, a $0.1\ \mu F$ capacitor and an inductor are connected in series across a $250\ V$ supply at variable frequency. Calculate the value of inductance of inductor at which resonance will occur. Given that the resonant frequency is $60\ \text{Hz}$.

- (A) $0.70\ \text{H}$
- (B) $70.3\ \text{mH}$
- (C) $7.03 \times 10^{-5}\ \text{H}$
- (D) $70.3\ \text{H}$

Q8: A $100\ \Omega$ resistance, a $0.1\ \mu F$ capacitor and an inductor are connected in series across a 250 V supply at variable frequency. Calculate the value of inductance of inductor at which resonance will occur. Given that the resonant frequency is 60 Hz.

- (A) 0.70 H
- (B) 70.3 mH
- (C) $7.03 \times 10^{-5} H$
- (D) 70.3 H

Solution:

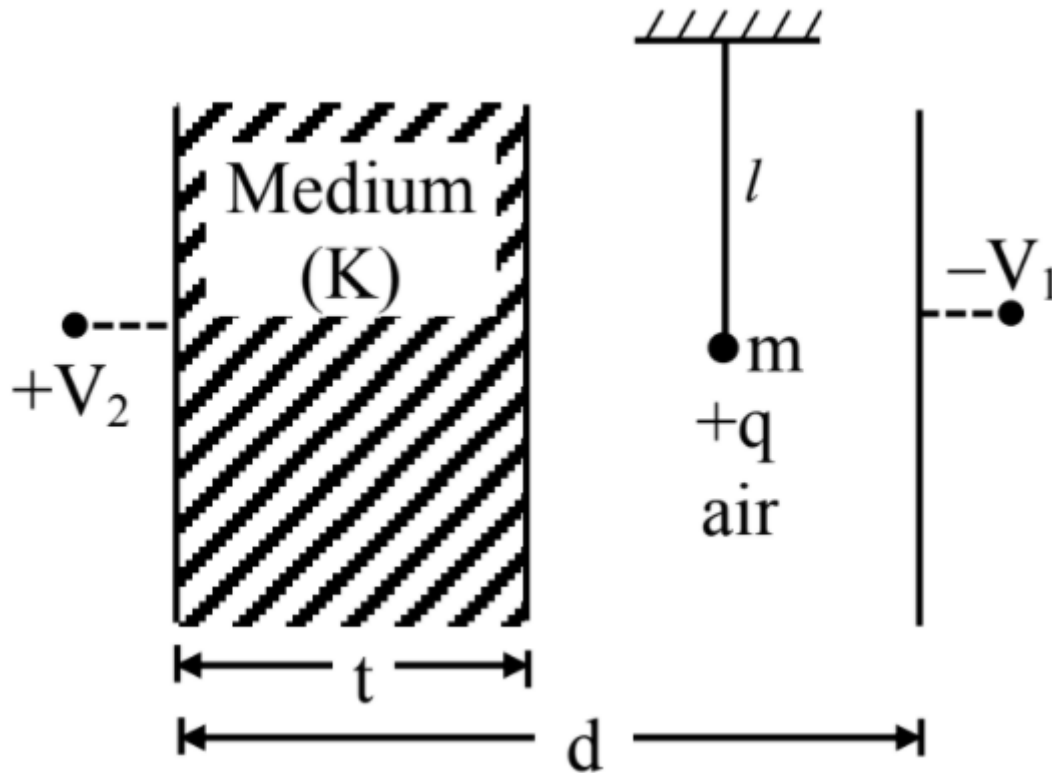
$$\omega = \frac{1}{\sqrt{LC}}$$

$$L = \frac{1}{\omega^2 C} = \frac{1}{[120\pi]^2 10^{-7}}$$

$$L = \frac{10^7}{120^2 \pi^2} \text{H}$$

$$= 70.3 \text{H}$$

Q9: A simple pendulum of mass 'm', length 'l' and charge '+ q' suspended in the electric field produced by two conducting parallel plates as shown. The capacitance of the Parallel plates with dielectric medium is C_2 and without dielectric medium is C_1 . The value of deflection of pendulum in equilibrium position will be



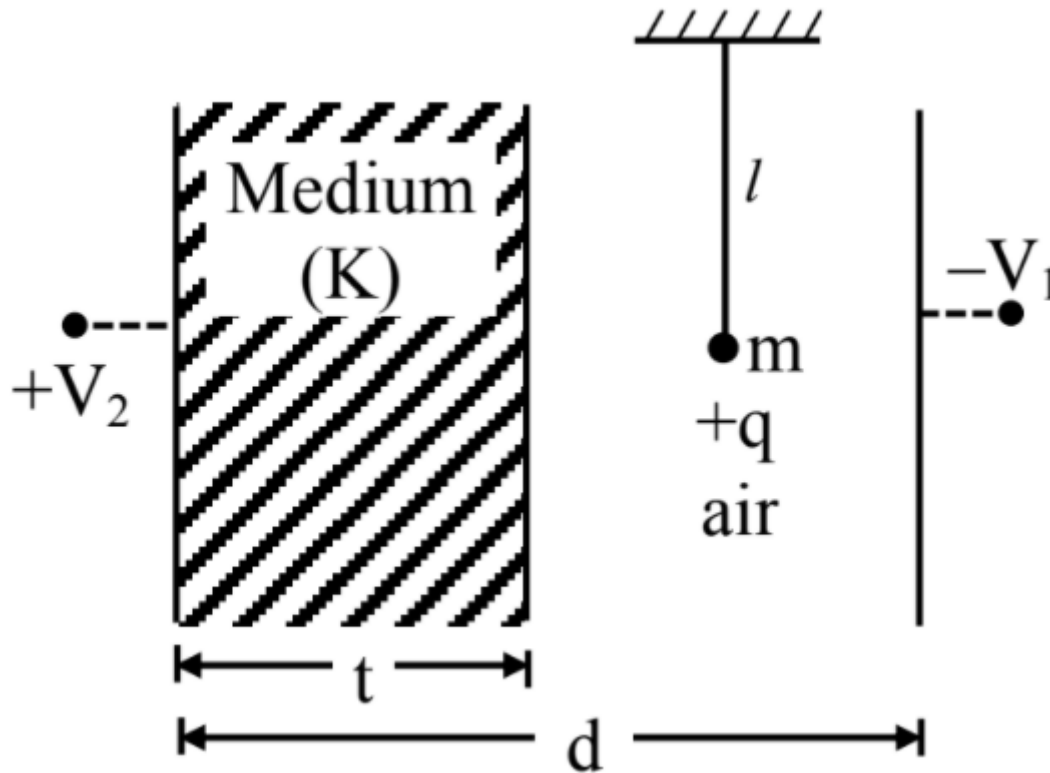
$$(A) \quad \tan^{-1} \left[\frac{q}{mg} \times \frac{C_1(V_2 - V_1)}{(C_1 + C_2)(d - t)} \right]$$

$$(B) \quad \tan^{-1} \left[\frac{q}{mg} \times \frac{C_2(V_2 - V_1)}{(C_1 + C_2)(d - t)} \right]$$

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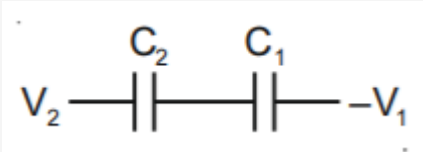
(B) $\tan^{-1} \left[\frac{q}{mg} \times \frac{C_2(V_2 - V_1)}{(C_1 + C_2)(d - t)} \right]$

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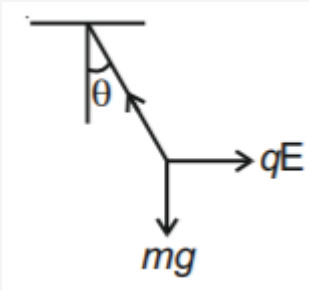
(D) $\tan^{-1} \left[\frac{q}{mg} \times \frac{C_1(V_1 + V_2)}{(C_1 + C_2)(d - t)} \right]$

Solution:

Potential across capacitor, $C_1 = \frac{C_2(V_1+V_2)}{(C_1+C_2)}$



Field in the region of $C_1 = \frac{C_2(V_1+V_2)}{(C_1+C_2)(d-t)}$



$$T \cos \theta = mg$$

$$T \sin \theta = qE$$

$$\tan \theta = \frac{qE}{mg}$$

$$\tan \theta = \frac{qC_2 (V_1 + V_2)}{(C_1 + C_2)(d - t)mg}$$

Q10: Two Carnot engines A and B operate in series such that engine A absorbs heat at T_1 and rejects heat to a sink at temperature T. Engine B absorbs half of the heat rejected by Engine A and rejects heat to the sink at T_3 . When work done in both the cases is equal, to value of T is:

(A) $\frac{2}{3}T_1 + \frac{3}{2}T_3$

(B) $\frac{1}{3}T_1 + \frac{2}{3}T_3$

(C) $\frac{3}{2}T_1 + \frac{1}{3}T_3$

(D) $\frac{2}{3}T_1 + \frac{1}{3}T_3$

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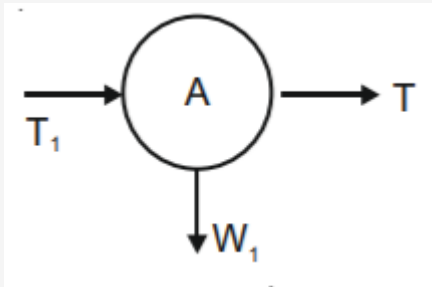
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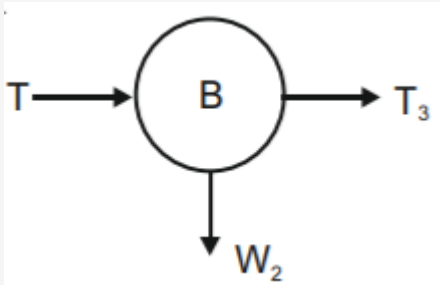
(C) $\frac{3}{2}T_1 + \frac{1}{3}T_3$

(D) $\frac{2}{3}T_1 + \frac{1}{3}T_3$

Solution:



$$W_1 = Q_1 - Q \Rightarrow \frac{W_1}{Q} = \frac{Q_1}{Q} - 1$$
$$= \frac{T_1}{T} - 1 \dots\dots\dots (i)$$



$$\Rightarrow \frac{2W_2}{Q} = 1 - \frac{T_3}{T} \dots\dots\dots (ii)$$

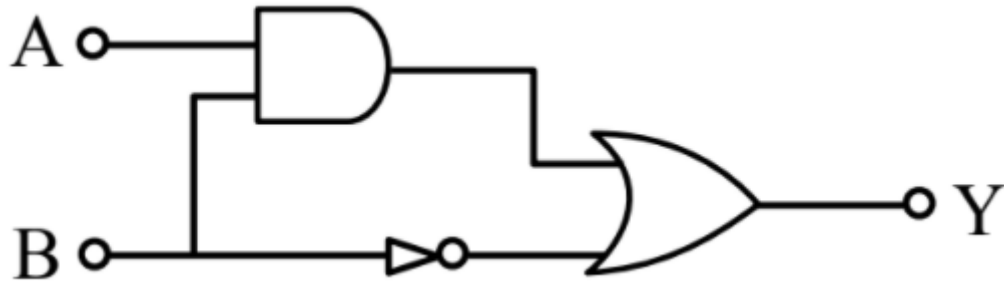
From (i) and (ii)

$$2 \left(\frac{T_1}{T} - 1 \right) = 1 - \frac{T_3}{T}$$

$$\Rightarrow \frac{2T_1}{T} + \frac{T_3}{T} = 3$$

$$\Rightarrow T = \frac{2T_1}{3} + \frac{T_3}{3}$$

Q11: Find the truth table for the function Y of A and B represented in the following figure.



(A)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

(B)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

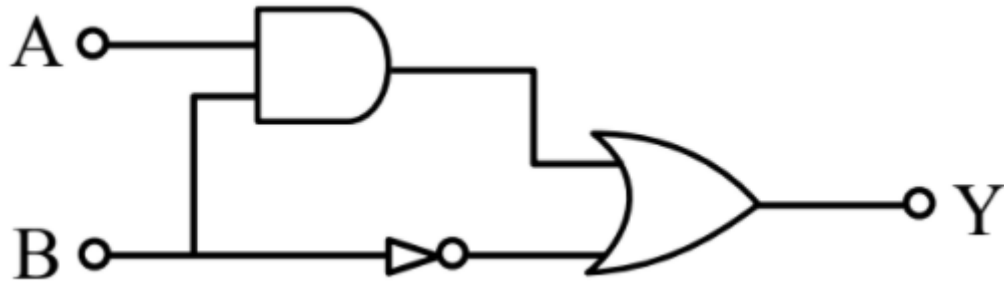
(C)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(D)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	0

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(C)

A	B	Y
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1	0	1
1	1	1

(D)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	0

Solution:

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

Q12: Figure A and B shown two long straight wires of circular cross-section (a and b with $a < b$), carrying current I which is uniformly distributed across the cross-section. The magnitude of magnetic field B varies with radius r and can be represented as:

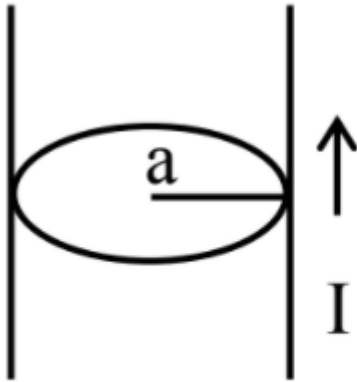


Fig. A

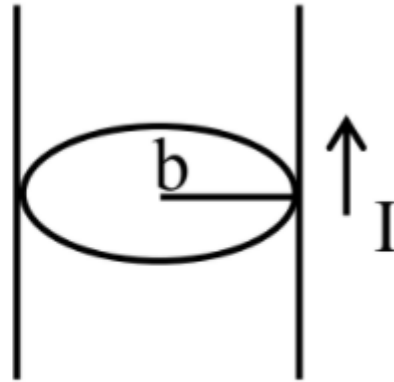
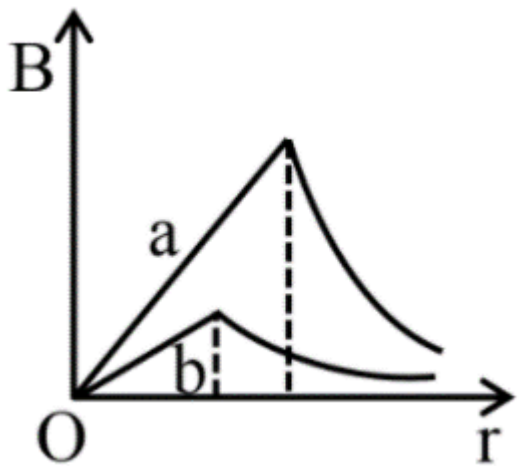
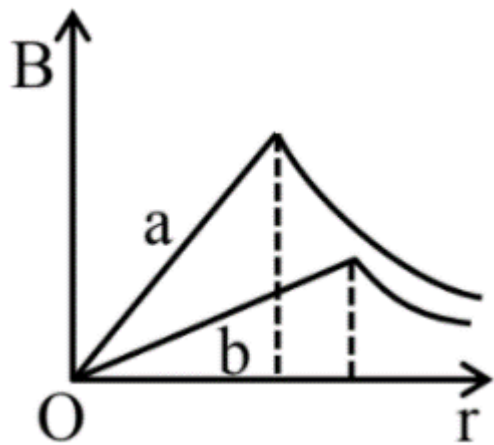


Fig. B

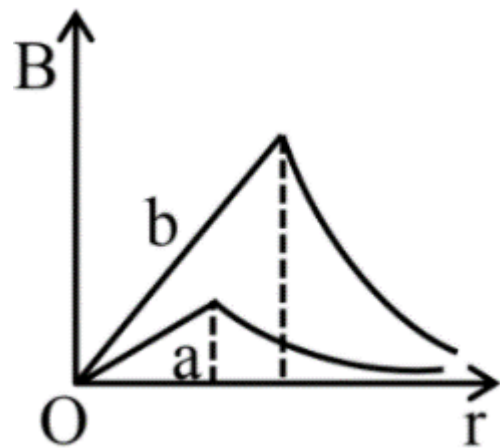
(A)



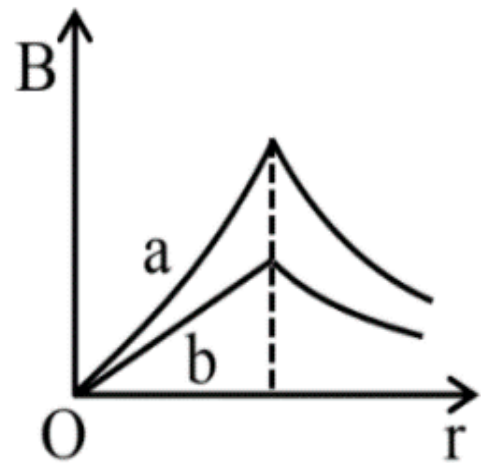
(B)



(C)



(D)



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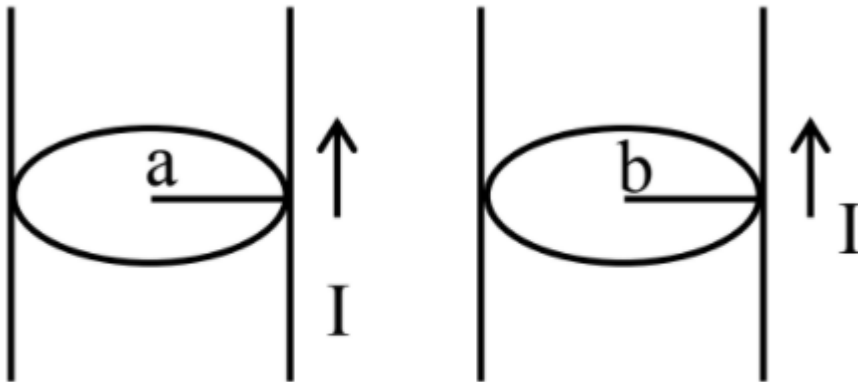
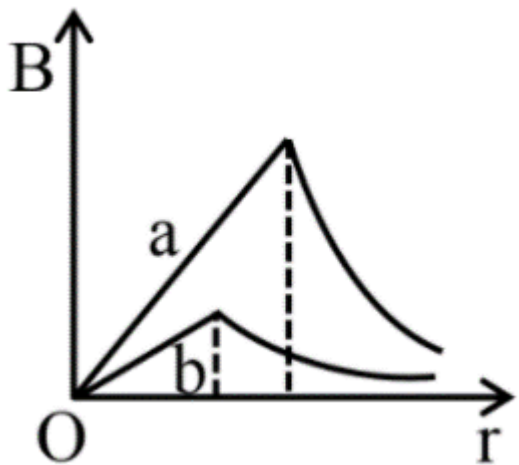


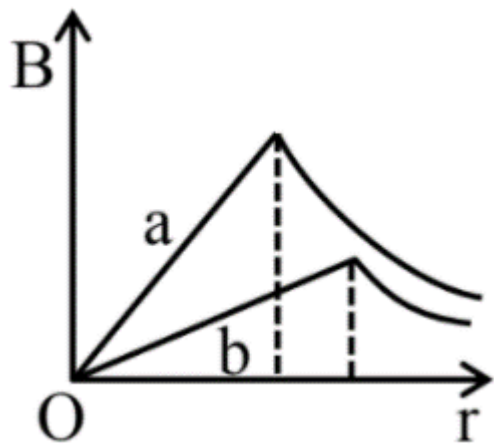
Fig. A

Fig. B

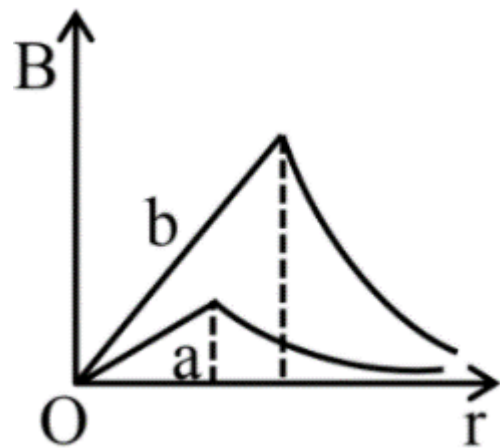
(A)



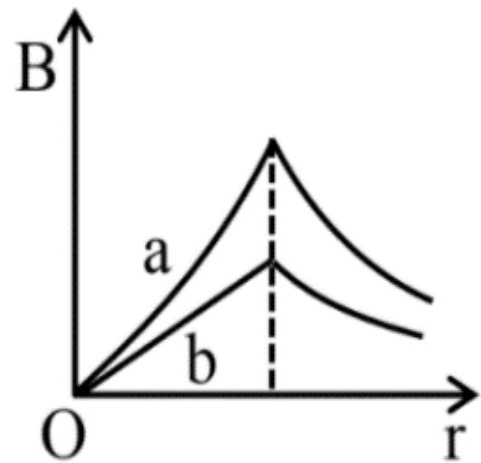
(B)



(C)



(D)



Solution:

For inside point

$$\mathbf{B} \cdot 2\pi r = \frac{\mu_0 I \cdot \pi r^2}{\pi a^2}$$

$$\mathbf{B} = \frac{\mu_0 I r}{2\pi a^2}$$

$$\text{For outside point } \mathbf{B} \cdot 2\pi r = \mu_0 I \quad \mathbf{B} = \frac{\mu_0 I}{2\pi r}$$

$$\text{For fig. A } \mathbf{B}(\text{at } r=a) = \frac{\mu_0 I}{2\pi a}$$

$$\text{For fig. B } \mathbf{B}(\text{at } r=b) = \frac{\mu_0 I}{2\pi b}$$

Option (b) is correct

Q13: Two identical particles of mass 1 kg each go round a circle of radius R, under the action of their mutual gravitational attraction. The angular speed of each particle is:

(A) $\sqrt{\frac{G}{2R^3}}$

(B) $\frac{1}{2} \sqrt{\frac{G}{R^3}}$

(C) $\frac{1}{2R} \sqrt{\frac{1}{G}}$

(D) $\sqrt{\frac{2G}{R^3}}$

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(D) $\sqrt{\frac{2G}{R^3}}$

Solution:

$$\frac{GM^2}{(2R)^2} = M\omega^2 R$$

$$\frac{GM}{4R^3} = \omega^2$$

$$\Rightarrow \omega = \frac{1}{2} \sqrt{\frac{GM}{R^3}}$$

$$= \frac{1}{2} \sqrt{\frac{G}{R^3}}$$

Q14: Consider the following statements:

A. Atoms of each element emit characteristics spectrum.

B. According to Bohr's Postulate, an electron in a hydrogen atom, revolves in a certain stationary orbit.

C. The density of nuclear matter depends on the size of the nucleus.

D. A free neutron is stable but a free proton decay is possible.

E. Radioactivity is an indication of the instability of nuclei.

Choose the correct answer from the options given below:

(A) A, B, C, D and E

(B) A, B and E only

(C) B and D only

(D) A, C and E only

Q14: Consider the following statements:

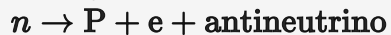
- A. Atoms of each element emit characteristics spectrum.
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Choose the correct answer from the options given below:

- (A) A, B, C, D and E
- ☒ (B) A, B and E only
- (C) B and D only
- (D) A, C and E only

Solution:

Density of nuclear matter is independent of size. Free neutron can decay

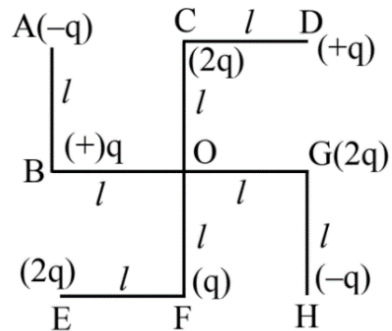


As $m_n > m_p + m_e$

So, A, B and E are correct

option b is correct

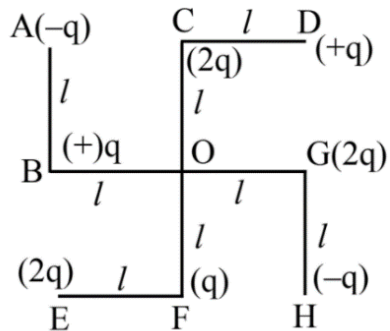
Q15: What will be the magnitude of electric field at point O as shown in figure? Each side of the figure is l and perpendicular to each other?



- (A) $\frac{1}{4\pi\epsilon_0} \frac{q}{l^2}$
- (B) $\frac{1}{4\pi\epsilon_0} \frac{q}{(2l^2)} (2\sqrt{2} - 1)$
- (C) $\frac{q}{4\pi\epsilon_0 (2l)^2}$

(D) $\frac{1}{4\pi\epsilon_0} \frac{2q}{2l^2} (\sqrt{2})$

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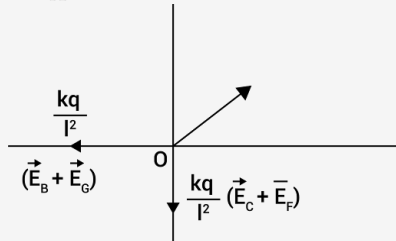
(C) $\frac{q}{4\pi\epsilon_0 (2l)^2}$

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Solution:

$$\vec{E}_0 = \vec{E}_A + \vec{E}_B + \vec{E}_C + \vec{E}_D + \vec{E}_E + \vec{E}_F + \vec{E}_G + \vec{E}_H$$

$$\vec{E}_A + \vec{E}_H = 0$$



$$\Rightarrow |\vec{E}_0| = \left| \frac{kq}{(\sqrt{2}l)^2} - \frac{kq}{l^2} \sqrt{2} \right|$$

$$= \frac{kq}{l^2} \left| \left(\frac{1}{2} - \sqrt{2} \right) \right|$$

$$= \frac{kq}{2l^2} (2\sqrt{2} - 1)$$

Q16: A physical quantity 'y' is represented by the formula $y = m^2 r^{-4} g^x l^{-\frac{3}{2}}$

If the percentage errors found in y, m, r, l and g are 18, 1, 0.5, 4 and p respectively, then find the value of x and p.

- (A) 5 and ± 2
- (B) 4 and ± 3
- (C) $\frac{16}{3}$ and $\pm \frac{3}{2}$
- (D) 8 and ± 2

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Solution:

$$\frac{\Delta y}{y} = \frac{2\Delta m}{m} + \frac{4\Delta r}{r} + \frac{x\Delta g}{g} + \frac{3}{2} \frac{\Delta l}{l}$$

$$18 = 2 \times 1 + 4 \times 0.5 + \frac{3}{2} \times 4 + px$$

$$8 = px$$

By checking from options, only option (c) is correct

Q17: An automobile of mass 'm' accelerates starting from origin and initially at rest, while the engine supplies constant power P. The position is given as a function of time by:

(A) $\left(\frac{9P}{8m}\right)^{\frac{1}{2}} t^{\frac{3}{2}}$

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Solution:

Power = P

So, K.E=Pt

$$\frac{1}{2}mv^2 = Pt$$

$$v^2 = \frac{2P}{m}t$$

$$v = \sqrt{\frac{2P}{m}}t^{\frac{1}{2}}$$

$$dx = \sqrt{\frac{2P}{m}}t^{\frac{1}{2}} dt$$

$$\int dx = \sqrt{\frac{2P}{m}} \int_0^t t^{\frac{1}{2}} dt$$

$$x = \sqrt{\frac{2P}{m}} \frac{t^{3/2}}{3/2}$$

$$x = \left(\frac{8P}{9m} \right)^{\frac{1}{2}} t^{3/2}$$

Q18: The planet Mars has two moons, if one of them has a period 7 hours, 30 minutes and an orbital radius of $9.0 \times 10^3 \text{ km}$. Find the mass of Mars.

$$\left\{ \text{Given } \frac{4\pi^2}{G} = 6 \times 10^{11} \text{ N}^{-1} \text{ m}^{-2} \text{ kg}^2 \right\}$$

- (A) $5.96 \times 10^{19} \text{ kg}$
- (B) $3.25 \times 10^{21} \text{ kg}$
- (C) $7.02 \times 10^{25} \text{ kg}$
- (D) $6.00 \times 10^{23} \text{ kg}$

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Solution:

$$m\omega^2 r = \frac{GMm}{r^2}$$

$$\omega^2 = \frac{GM}{r^3}$$

$$\frac{4\pi^2}{T^2} = \frac{GM}{r^3}$$

$$T^2 = \frac{4\pi^2 r^3}{GM} \quad T = 2\pi \sqrt{\frac{r^3}{GM}}$$

$$M = \frac{4\pi^2 r^3}{GT^2} = \frac{6 \times 10^{11} \times 729 \times 10^{18}}{T^2}$$

$$M = 6.0 \times 10^{23} \text{ kg}$$

Q19: A particle of mass M originally at rest is subjected to a force whose direction is constant but magnitude varies with time according to the relation

$$F = F_0 \left[1 - \left(\frac{t-T}{T} \right)^2 \right]$$

Where F_0 and T are constants. The force acts only for the time interval $2T$. The velocity v of the particle after time $2T$ is:

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Solution:

$$F = F_0 \left[1 - \left(\frac{t - T}{T} \right)^2 \right]$$

$$a = \frac{F_0}{M} \left[1 - \left(\frac{t - T}{T} \right)^2 \right]$$

$$\Rightarrow \int_0^v dv = \int_0^{2T} \frac{F_0}{M \times T^2} [T^2 - (t^2 + T^2 - 2tT)] dt$$

$$\Rightarrow V = \frac{F_0}{MT^2} \times \left[2T \times \frac{t^2}{2} - \frac{t^3}{3} \right]_0^{2T}$$

$$= \frac{F_0}{MT^2} \times \left[T \times 4T^2 - \frac{1}{3} \times 8T^3 \right]$$

$$= \frac{F_0}{MT^2} \times \frac{4T^3}{3} = \frac{4F_0T}{3M}$$

Q20: The resistance of a conductor at 15°C is $16\ \Omega$ and at 100°C is $20\ \Omega$. What will be the temperature coefficient of resistance of the conductor?

- (A) $0.010^{\circ}\text{C}^{-1}$
- (B) $0.033^{\circ}\text{C}^{-1}$
- (C) $0.003^{\circ}\text{C}^{-1}$
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Solution:

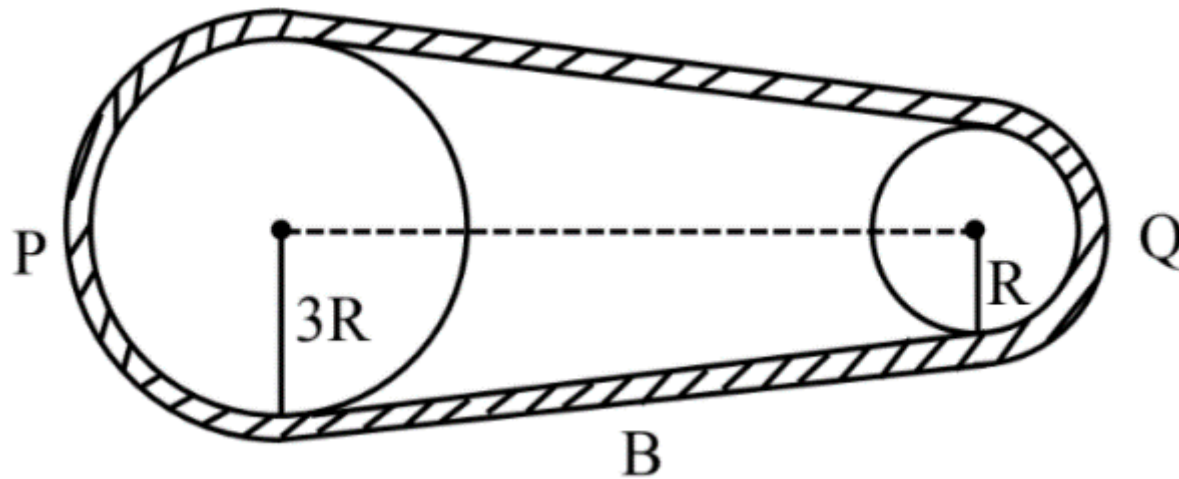
$$R_t = R_0[1 + \alpha(\Delta T)]$$

$$20 = 16[1 + \alpha(85)]$$

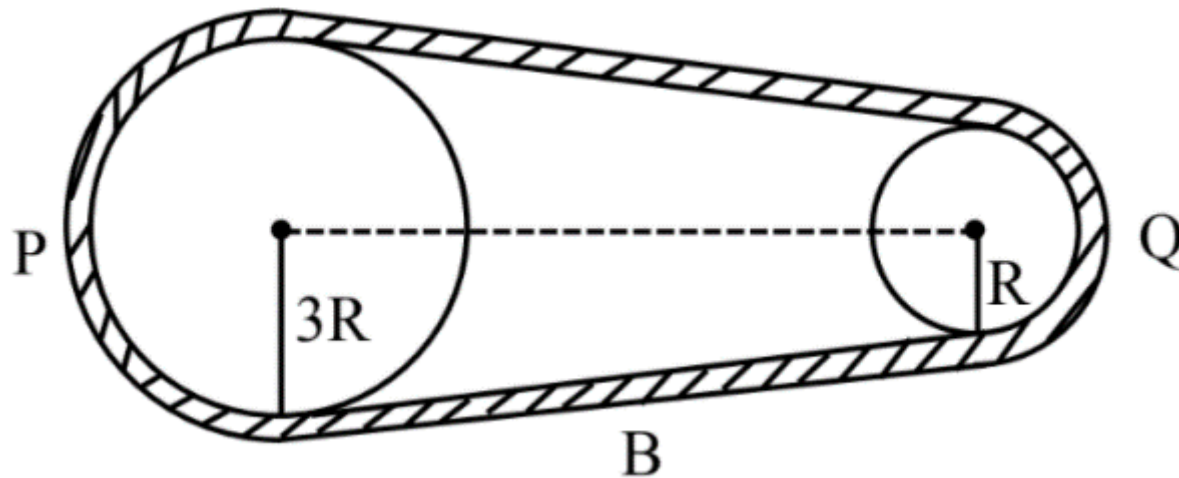
$$\alpha = \frac{4}{16 \times 85}$$

$$= 0.003^{\circ}\text{C}^{-1}$$

Q21: In the given figure, two wheels P and Q are connected by a belt B. The radius of P is three times as that of Q. In case of same rotational kinetic energy, the ratio of rotational inertias $\left(\frac{I_1}{I_2}\right)$ will be $x:1$. The value of x will be _____.



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9.00

Solution:

$$\frac{1}{2}I_1\omega_1^2 = \frac{1}{2}I_2\omega_2^2$$

$$\frac{1}{2}I_1\left(\frac{v}{R_1}\right)^2 = \frac{1}{2}I_2\left(\frac{v}{R_2}\right)^2$$

$$\frac{I_1}{I_2} = \left(\frac{R_1}{R_2}\right)^2 = \frac{9}{1}$$

Q22: The difference in the number of waves when yellow light propagates through air and vacuum columns of the same thickness is one. The thickness of the air column is _____ mm. [Refractive index of air = 1.0003, wavelength of yellow light in vacuum = $6000 \text{ \AA}$]

Q22: The difference in the number of waves when yellow light propagates through air and vacuum columns of the same thickness is one. The thickness of the air column is _____ mm. [Refractive index of air = 1.0003, wavelength of yellow light in vacuum = $6000 \text{ \AA}$]

2.00

Solution:

$$\Delta n = \frac{d}{\lambda_1} - \frac{d}{\lambda_2}$$

$$1 = \frac{(\mu - 1)d}{\lambda_0}$$

$$\Rightarrow d = 6 \times 10^{-7} / 0.0003$$

$$= 2 \times 10^{-3} \text{ m}$$

Q23: The maximum amplitude for an amplitude modulated wave is found to be 12V while the minimum amplitude is found to be 3V. The modulation index is $0.6x$ where x _____.

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Solution:

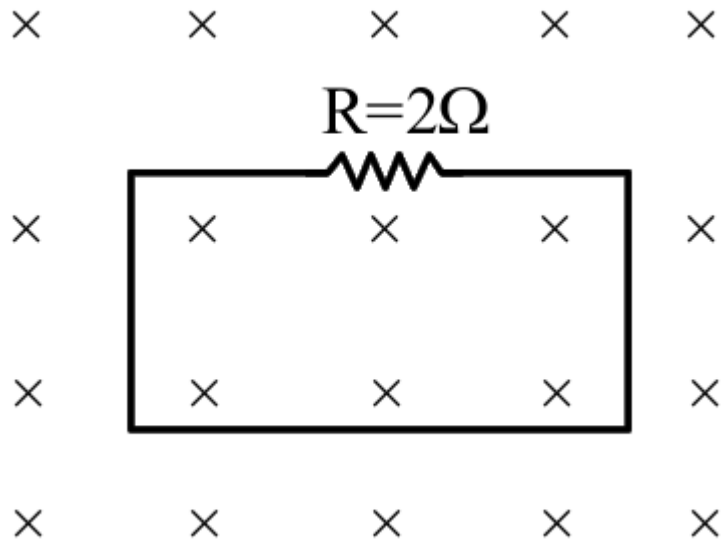
$$M + A = 12$$

$$A - M = 3$$

$$\Rightarrow A = \frac{15}{2} \text{ and } M = \frac{9}{2}$$

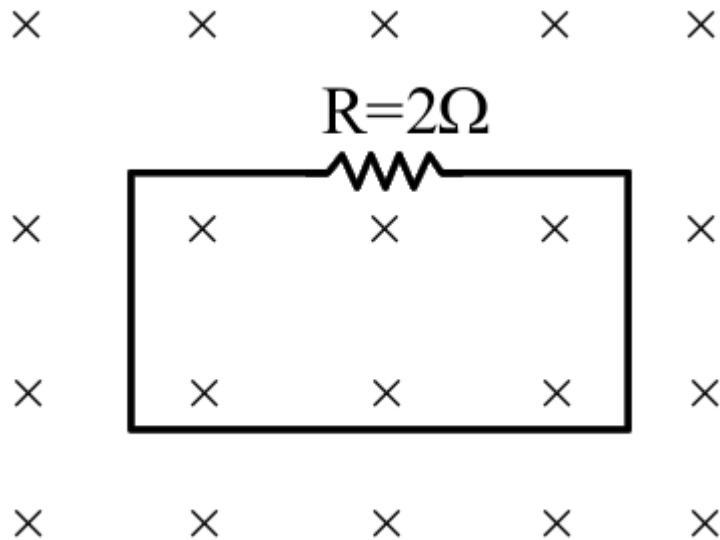
$$m = \frac{M}{A} = \frac{9}{15} = 0.6$$

Q24: If the given figure the magnetic flux through the loop increases according to the relation $\phi_B(t) = 10t^2 + 20t$, where ϕ_B is in milliwebers and t is in seconds. The magnitude of current through $R = 2\Omega$ resistor at $t = 5$ s is _____ mA.





Q24: If the given figure the magnetic flux through the loop increases according to the relation $\phi_B(t) = 10t^2 + 20t$, where ϕ_B is in milliwebers and t is in seconds. The magnitude of current through $R = 2\Omega$ resistor at $t = 5$ s is _____ mA.



60.00

Solution:

$$\varepsilon = \left| \frac{d\phi}{dt} \right| = (20t + 20) \times 10^{-3}$$

$$i = \frac{\varepsilon}{R} = \frac{(20 \times 5 + 20)}{2} \times 10^{-3} = 60 \text{ mA}$$

Q25: A particle executes simple harmonic motion represented by displacement function as

$$x(t) = A \sin(\omega t + \phi)$$

If the position and velocity of the particle at $t = 0$ s are 2 cm and $2\omega \text{ cm s}^{-1}$ respectively, then its amplitude is $x\sqrt{2} \text{ cm}$ where the value of x is _____.

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Solution:

$$|A \sin \phi| = 2$$

$$|\omega A \cos \phi| = 2\omega$$

$$\Rightarrow |\tan \phi| = 1$$

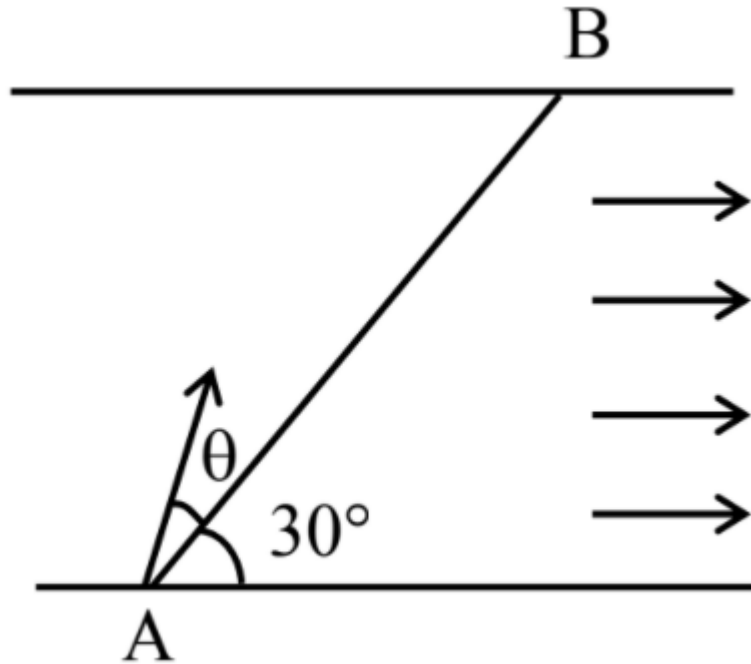
$$\Rightarrow \phi = \frac{\pi}{4}$$

$$x(0) = A \sin\left(\frac{\pi}{4}\right) = 2 \text{ cm}$$

$$\Rightarrow A = 2\sqrt{2} \text{ cm}$$

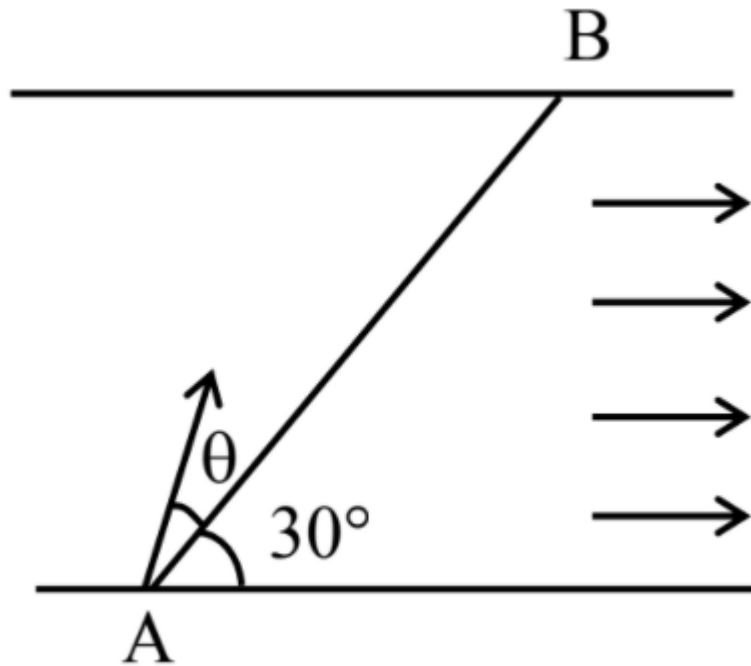
$$\Rightarrow x = 2$$

Q26: A swimmer wants to cross a river from point A to point B. Line AB makes an angle of 30° with the flow of river. Magnitude of velocity of the swimmer is same as that of the river. The angle θ with the line AB should be _____ $^\circ$, so that the swimmer reaches point B.



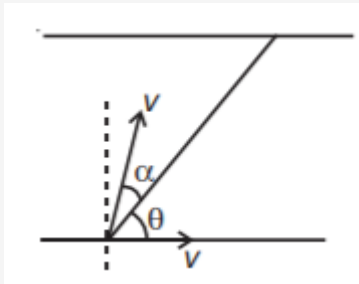


Q26: A swimmer wants to cross a river from point A to point B. Line AB makes an angle of 30° with the flow of river. Magnitude of velocity of the swimmer is same as that of the river. The angle θ with the line AB should be _____ $^\circ$, so that the swimmer reaches point B.



30.00

Solution:



For equal magnitude v_s and v_R

$$\alpha = \theta$$

$$\Rightarrow \alpha = 30^\circ$$

Q27: For the circuit shown, the value of current at time $t = 3.2$ s will be _____ A.

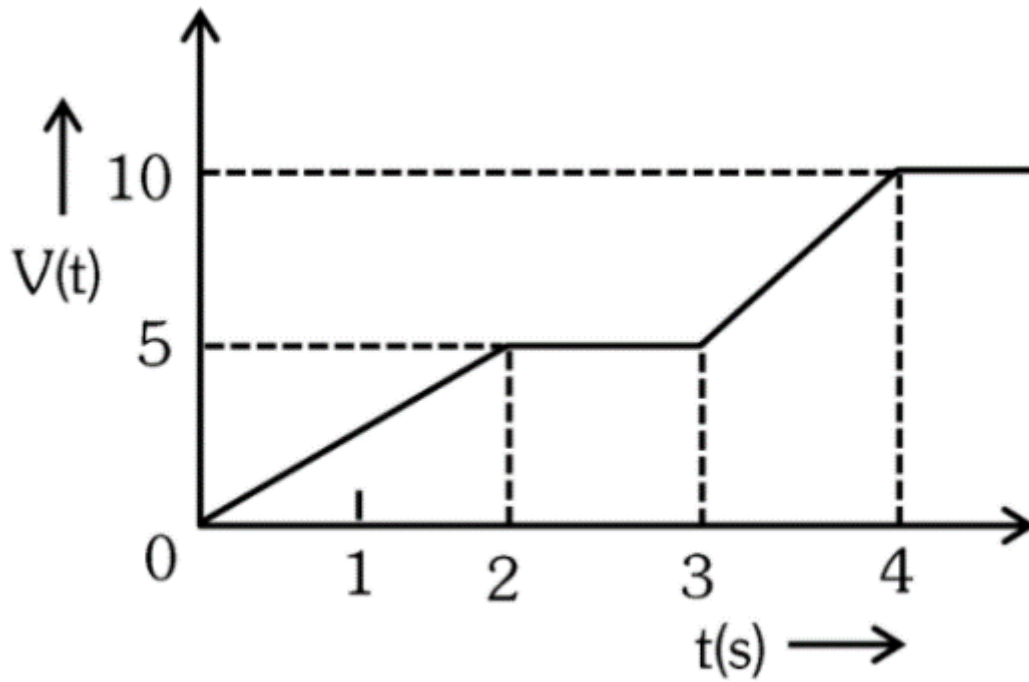


Figure 1

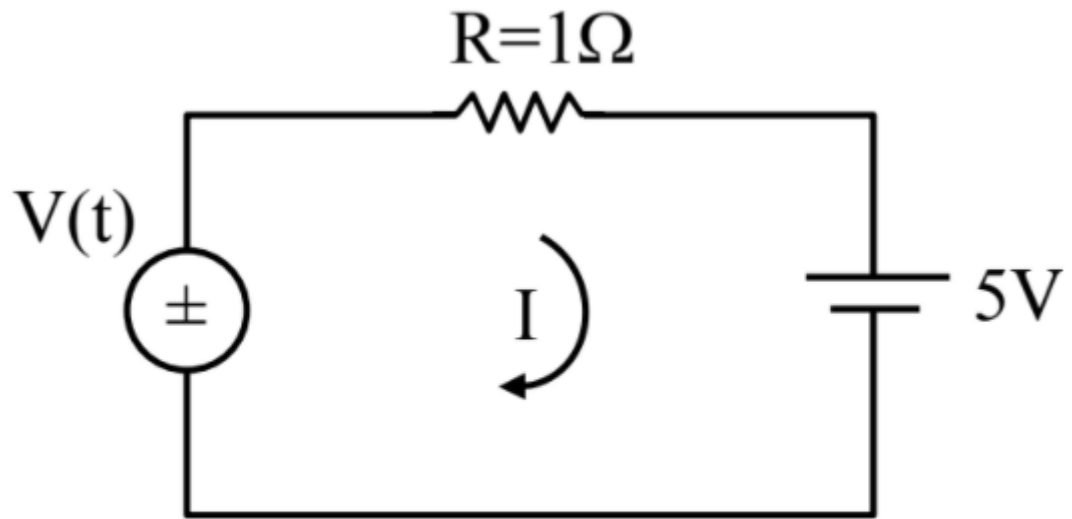


Figure-2

[Voltage distribution $V(t)$ is shown by Fig. (1) and the circuit is shown in Fig. (2)]



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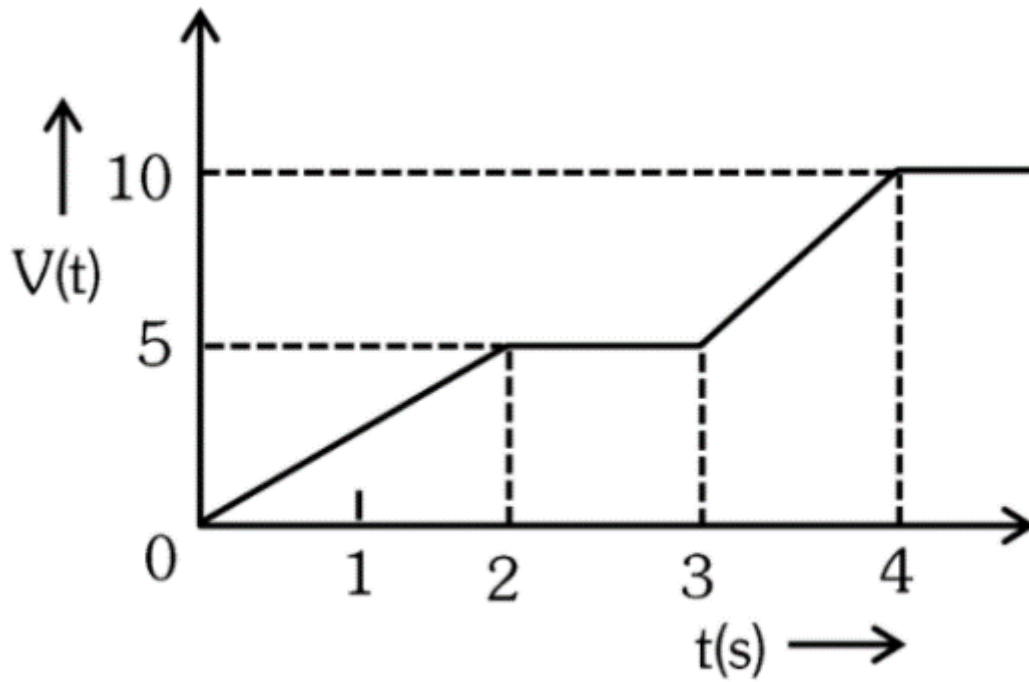


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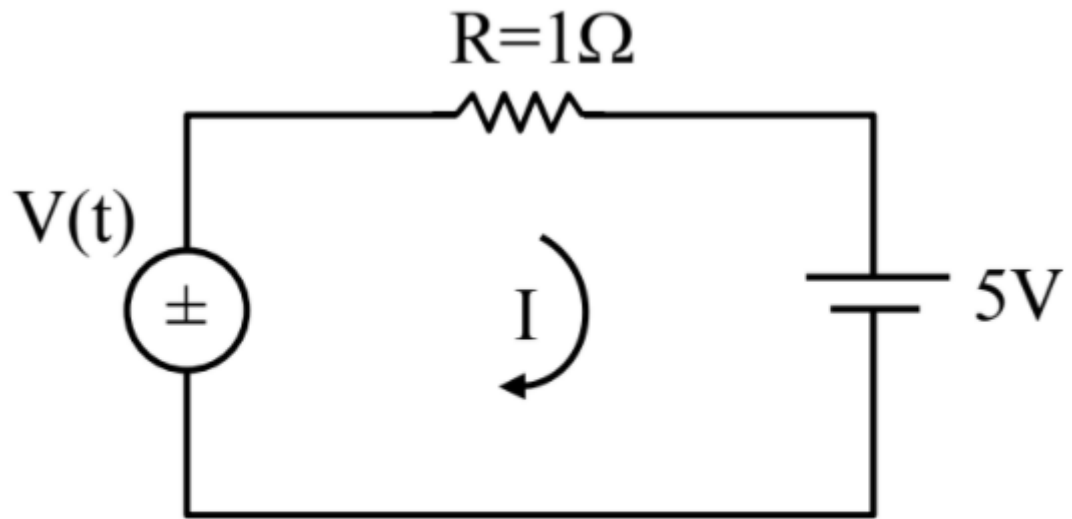


Figure-2

[Voltage distribution $V(t)$ is shown by Fig. (1) and the circuit is shown in Fig. (2)]

1.00

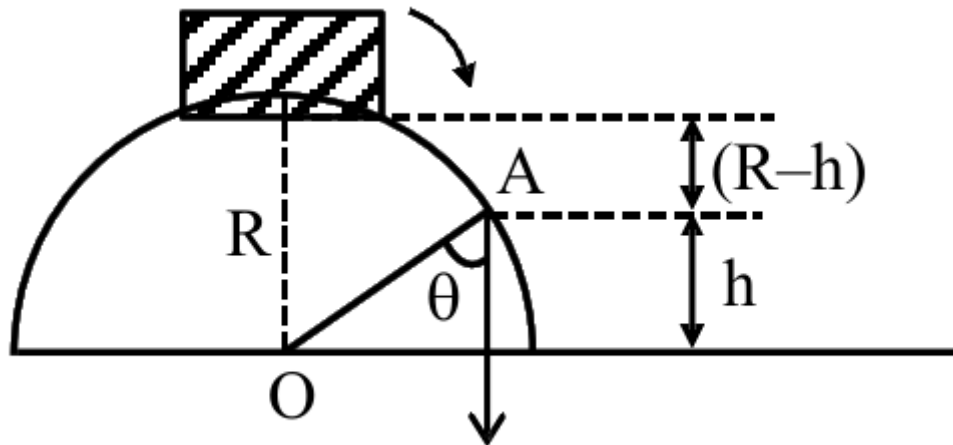
Solution:

$$V_{(t=3.2)} = 6 \text{ V}$$

$$I = \frac{6 - 5}{1} \text{ A} = 1 \text{ A}$$

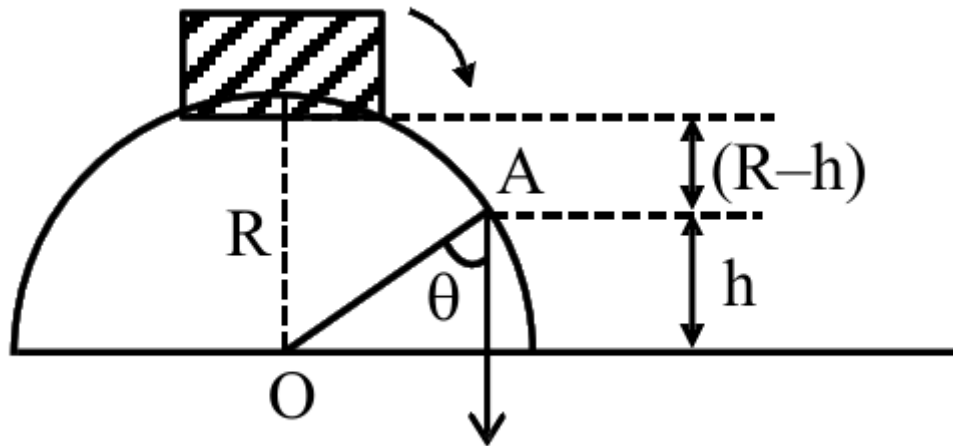
Q28: A small block slides down from the top of hemisphere of radius $R = 3$ m as shown in the figure. The height ' h ' at which the block will lose contact with the surface of the sphere is _____m.

(Assume there is no friction between the block and the hemisphere)



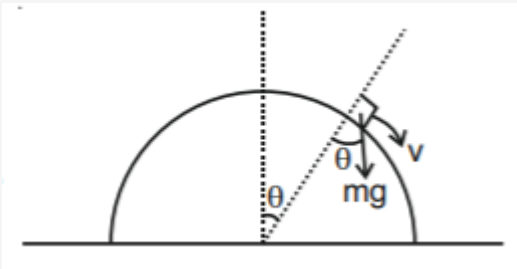
Q28: A small block slides down from the top of hemisphere of radius $R = 3$ m as shown in the figure. The height ' h ' at which the block will lose contact with the surface of the sphere is _____m.

(Assume there is no friction between the block and the hemisphere)



2.00

Solution:



$$mg \cos \theta = \frac{mv^2}{R}$$

$$\frac{1}{2}mv^2 = mgR(1 - \cos \theta)$$

On solving $\cos \theta = \frac{2}{3}$

$$h = \frac{2R}{3} = 2 \text{ m}$$

Q29: The K_{α} X-ray of molybdenum has wavelength 0.071 nm. If the energy of a molybdenum atoms with a K electron knocked out is 27.5 keV, the energy of this atom when an L electron is knocked out will be _____ keV. (Round off to the nearest integer)

$$[h = 4.14 \times 10^{-15} \text{ eVs}, c = 3 \times 10^8 \text{ ms}^{-1}]$$

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$$[h = 4.14 \times 10^{-15} \text{ eVs}, c = 3 \times 10^8 \text{ ms}^{-1}]$$

Solution:

$$\lambda_{K_\alpha} = \frac{hc}{E_K - E_L}$$

$$E_L = E_K - \frac{hc}{\lambda_{K_\alpha}}$$

$$= 27.5 \times 10^3 - \frac{4.14 \times 10^{-15} \times 3 \times 10^8}{0.071 \times 10^{-9}} \text{ eV}$$

$$= 27.5 - 17.49 \text{ KeV}$$

$$\approx 10 \text{ KeV}$$

Q30: The water is filled up to height of 12 m in a tank having vertical sidewalls. A hole is made in one of the walls at a depth 'h' below the water level. The value of 'h' for which the emerging stream of water strikes the ground at the maximum range is ____ m.

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Solution:

For maximum range

$$h = H - h$$

$$h = \frac{H}{2}$$

$$= \frac{12}{2} \text{ m} = 6 \text{ m}$$

Q31: Which one of the following set of elements can be detected using sodium fusion extract?

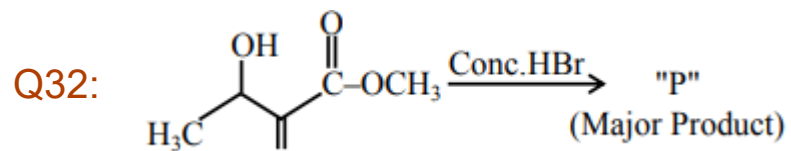
- (A) Sulfur, Nitrogen, Phosphorous, Halogens
- (B) Phosphorous, Oxygen, Nitrogen, Halogens
- (C) Nitrogen, Phosphorous, Carbon, Sulfur
- (D) Halogens, Nitrogen, Oxygen, Sulfur

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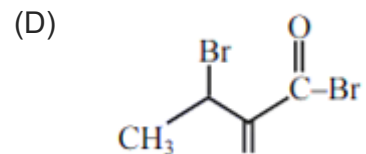
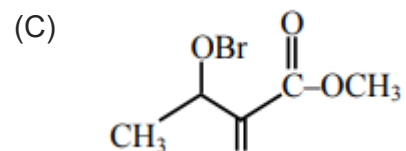
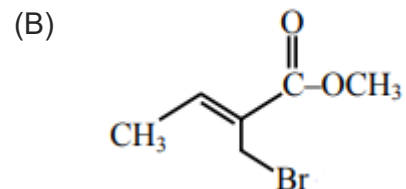
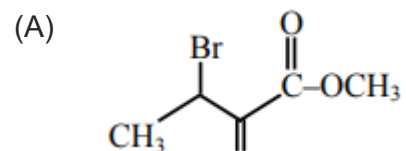
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- (C) Nitrogen, Phosphorous, Carbon, Sulfur
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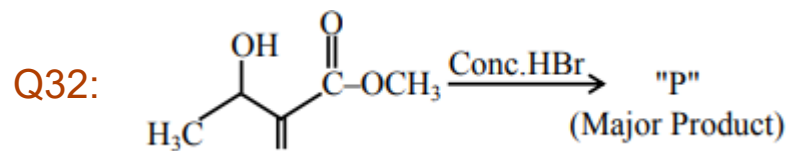
Solution:

By sodium fusion extract we can detect sulphur, nitrogen, Phosphorous and halogens, because they are converted in to their ionic form with sodium metal.

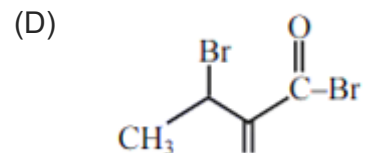
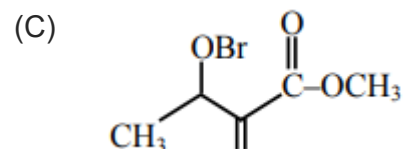
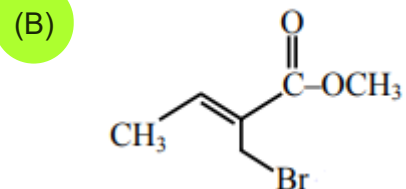
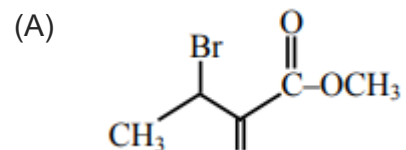


Consider the above reaction, the major product "P" formed is:

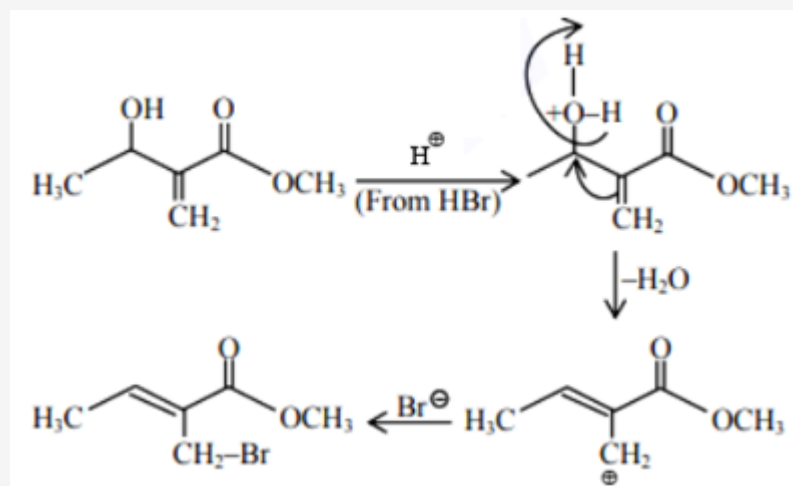




Consider the above reaction, the major product "P" formed is:



Solution:



Q33: The number of neutrons and electrons, respectively, present in the radioactive isotope of hydrogen is

- (A) 1 and 1
- (B) 3 and 1
- (C) 2 and 1
- (D) 2 and 2

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- (A) 1 and 1
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Solution:

Radioactive isotope of Hydrogen is tritium (${}^3_1\text{H}$)

Number of p = 1, n = 2, e^- = 1.

Q34: Match **List-I** with **List-II**.

List-I	List-II
(a) Li	(i) photoelectric cell
(b) Na	(ii) absorbent of CO ₂
(c) K	(iii) coolant in fast breeder nuclear reactor
(d) Cs	(iv) treatment of cancer
	(v) bearings for motor engines

Choose the **correct** answer from the options given below :

- (A) (a) - (v), (b) - (i), (c) - (ii), (d) - (iv)
- (B) (a) - (v), (b) - (ii), (c) - (iv), (d) - (i)
- (C) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)
- (D) (a) - (v), (b) - (iii), (c) - (ii), (d) - (i)

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(C) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

(D) (a) - (v), (b) - (iii), (c) - (ii), (d) - (i)

Solution:

- (a) Lithium metal is used to make useful alloys, for example with lead to make 'white metal' bearings for motor engines.
- (b) Liquid sodium metal is used as a coolant in fast breeder nuclear reactors.
- (c) Potassium hydroxide is used in the manufacture of soft soap. It is also used as an excellent absorbent of carbon dioxide.
- (d) Caesium is used in devising photoelectric cells.

Q35: Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: $\text{SO}_2(\text{g})$ is adsorbed to a large extent than $\text{H}_2(\text{g})$ on activated charcoal.

Reason R: $\text{SO}_2(\text{g})$ has a higher critical temperature than $\text{H}_2(\text{g})$.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Both **A** and **R** are correct, but **R** is not the correct explanation **A**
- (B) Both **A** and **R** are correct, and **R** is the correct explanation of **A**.
- (C) A is not correct but R is correct.
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- ☒ (B) Both **A** and **R** are correct, and **R** is the correct explanation of **A**.
- (C) A is not correct but R is correct.
- (D) A is correct but R is not correct.

Solution:

In general, easily liquefiable gases (i.e., with higher critical temperatures) are readily adsorbed as van der Waals' forces are stronger near the critical temperatures. Thus, activated charcoal adsorbs more sulphur dioxide (critical temperature 630 K), than dihydrogen (critical temperature 33 K).

Q36: The **CORRECT** order of first ionisation enthalpy is:

- (A) $\text{Mg} < \text{S} < \text{Al} < \text{P}$
- (B) $\text{Mg} < \text{Al} < \text{S} < \text{P}$
- (C) $\text{Al} < \text{Mg} < \text{S} < \text{P}$
- (D) $\text{Mg} < \text{Al} < \text{P} < \text{S}$

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- (D) $\text{Mg} < \text{Al} < \text{P} < \text{S}$

Solution:

In general, the ionisation enthalpy increases from left to right in the periodic table.

Due to fully filled $3s^2$ configuration, the first ionisation energy of $\text{Mg} > \text{Al}$.

Due to half filled $3p^3$ configuration, the first ionisation energy of $\text{P} > \text{S}$.

Q37: Given below are two statements:

Statement I: Hyperconjugation is a permanent effect.

Statement II: Hyperconjugation in ethyl cation $\left(\text{CH}_3 - \overset{+}{\text{C}}\text{H}_2\right)$ involves the overlapping of $\text{C}_{\text{sp}^2} - \text{H}_{1\text{s}}$ bond with empty 2p orbital of other carbon.

Choose the correct option:

- (A) Both **statement I** and **statement II** are false
- (B) **Statement I** is incorrect, but **statement II** is true
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Solution:

Hyperconjugation is permanent effect.

$\left(\text{CH}_3 - \overset{+}{\text{C}}\text{H}_2\right)$ involve $\text{C}_{\text{sp}^3} - \text{H}_{1\text{s}}$ bond with empty 2p orbital hence given statement is false.

Q38: Given below are two statements :

Statement I: $[\text{Mn}(\text{CN})_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ are d^2sp^3 hybridized.

Statement II: $[\text{MnCl}_6]^{3-}$ and $[\text{FeF}_6]^{3-}$ are paramagnetic and have 4 and 5 unpaired electrons, respectively.

In the light of the above statements, choose the correct answer from the options given below:

- (A) **Statement I** is correct, but **statement II** is false
- (B) Both **statement I** and **statement II** are false
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(C) **Statement I** is incorrect, but **statement II** is true

(D) Both **statement I** and **statement II** are true

Solution:

$[Mn(CN)_6]^{3-} : Mn^{3+} = [Ar] 3d^4$. In presence of strong field ligand, the electrons are paired against Hund's rule and the hybridization is d^2sp^3 .

$[Fe(CN)_6]^{3-} : Fe^{3+} = [Ar] 3d^5$. In presence of strong field ligand, the electrons are paired against Hund's rule and the hybridization is d^2sp^3 .

$[Co(C_2O_4)_3]^{3-} : Co^{3+} = [Ar] 3d^6$. In presence of strong field ligand, the electrons are paired against Hund's rule and the hybridization is d^2sp^3 .

$[MnCl_6]^{3-} : Mn^{3+} = [Ar] 3d^4$. In presence of weak field ligand, the electrons are not paired against Hund's rule and the number of unpaired electrons = 4.

$[FeF_6]^{3-} : Fe^{3+} = [Ar] 3d^5$. In presence of weak field ligand, the electrons are not paired against Hund's rule and the number of unpaired electrons = 5.

Q39: To an aqueous solution containing ions such as Al^{3+} , Zn^{2+} , Ca^{2+} , Fe^{3+} , Ni^{2+} , Ba^{2+} and Cu^{2+} was added conc. HCl , followed by H_2S . The total number of cations precipitated during this reaction is/are:

- (A) 1
- (B) 3
- (C) 4
- (D) 2

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(A) 1

(B) 3

(C) 4

(D) 2

Solution:

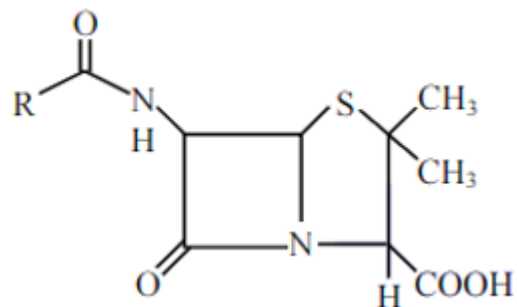
Group II and Group I cations except Ag^+ gives precipitate with dil. HCl + H_2S .

$\left\{ \begin{array}{l} \text{2}^{\text{nd}} \text{ group cation } \text{Cu}^{2+} \\ \text{3}^{\text{rd}} \text{ group cation } \text{Al}^{3+}, \text{Fe}^{3+} \\ \text{4}^{\text{th}} \text{ group cation } \text{Ni}^{2+}, \text{Zn}^{2+} \\ \text{5}^{\text{th}} \text{ group cation } \text{Ba}^{2+}, \text{Ca}^{2+} \end{array} \right.$

Q40: Given below are two statements:

Statement I: Penicillin is a bacteriostatic type of antibiotic.

Statement II: The general structure of Penicillin is:



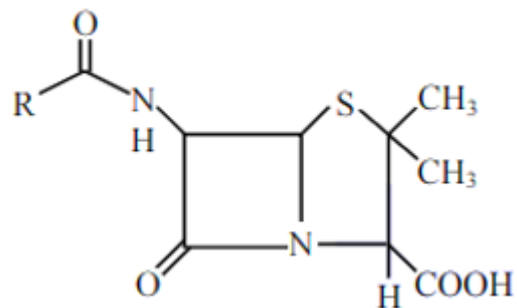
Choose the correct option:

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Choose the correct option:

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- ☒ (B) **Statement I** is incorrect, but **statement II** is true
- (C) Both **statement I** and **statement II** are true
- (D) **Statement I** is correct, but **statement II** is false

Solution:

Penicillin is bactericidal not bacteriostatic hence given statement is false. While it is the correct structure of Penicillin.

Q41: Compound A gives D-Galactose and D-Glucose on hydrolysis. The compound A is:

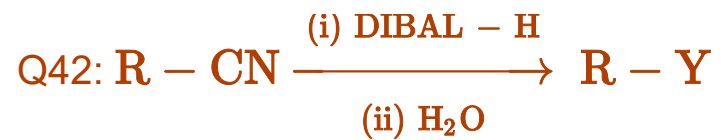
- (A) Amylose
- (B) Sucrose
- (C) Maltose
- (D) Lactose

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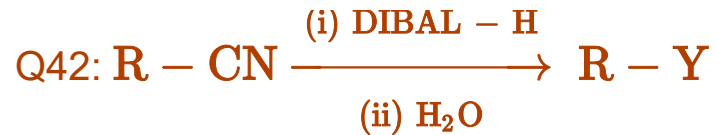
Solution:

Lactose is a disaccharide of β -D-Galactose and β -D-Glucose with C₁ of galactose and C₄ of glucose link.



Consider the above reaction and identify "Y".

- (A) $-\text{CH}_2\text{NH}_2$
- (B) $-\text{CONH}_2$
- (C) $-\text{CHO}$
- (D) $-\text{COOH}$



Consider the above reaction and identify "Y".

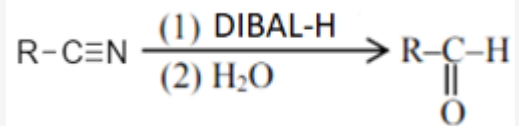
(A) $-\text{CH}_2\text{NH}_2$

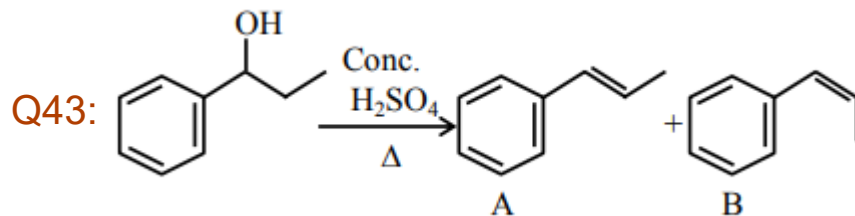
(B) $-\text{CONH}_2$

(C) $-\text{CHO}$

(D) $-\text{COOH}$

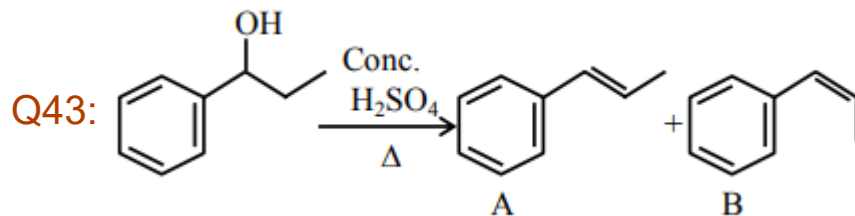
Solution:





consider the above reaction, and choose the correct statement:

- (A) The reaction is not possible in acidic medium
- (B) Both compounds A and B are formed equally
- (C) Compound A will be the major product
- (D) Compound B will be the major product



consider the above reaction, and choose the correct statement:

- (A) The reaction is not possible in acidic medium
- (B) Both compounds A and B are formed equally
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- (D) Compound B will be the major product

Solution:

Due to steric crowding trans alkene is more stable.

Q44: Match **List – I** with **List-II**.

	List-I		List-II
(a)	Carbon monoxide	(i)	Carcinogenic
(b)	Sulphur dioxide	(ii)	Metabolized by pyrus plants
(c)	Polychlorinated biphenyls	(iii)	Haemoglobin
(d)	Oxides of Nitrogen	(iv)	Stiffness of flower buds

Choose the correct answer from the options given below:

- (A) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)
- (B) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)
- (C) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)
- (D) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

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(d)	Oxides of Nitrogen	(iv)	Stiffness of flower buds

Choose the correct answer from the options given below:

(A) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

(B) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

(C) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

(D) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

Solution:

Carbon monoxide – Haemoglobin

Sulphur dioxide – Stiffness of flower buds

Polychlorinated biphenyls – Carcinogenic

Oxides of Nitrogen – Metabolized by pyrus plants

Q45: If the Thomson model of the atom was correct, then the result of Rutherford's gold foil experiment would have been:

- (A) All of the α -particles pass through the gold foil without decrease in speed.
- (B) α -Particles are deflected over a wide range of angles.
- (C) All α -particles get bounced back by 180°
- (D) α -Particles pass through the gold foil deflected by small angles and with reduced speed.

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- (D) α -Particles pass through the gold foil deflected by small angles and with reduced speed.

Solution:

Thomson's model states that atom has positive charge distributed throughout the sphere with electrons embedded in it at some places. Thus, alpha particles, which are positively charged would have been repelled by the positive charge of the atom and thus would have been deflected by small angles and with reduced speed.

Q46: Number of Cl = O bonds in chlorous acid, chloric acid and perchloric acid respectively are:

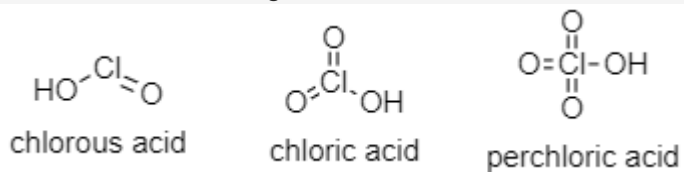
- (A) 3, 1 and 1
- (B) 4, 1 and 0
- (C) 1, 1 and 3
- (D) 1, 2 and 3

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- (A) 3, 1 and 1
- (B) 4, 1 and 0
- (C) 1, 1 and 3
- (D) 1, 2 and 3

Solution:

The structures are as given below –



Q47: Select the correct statements.

(A) Crystalline solids have long range order.

(B) Crystalline solids are isotropic.

(C) Amorphous solid are sometimes called pseudo solids.

(D) Amorphous solids soften over a range of temperatures.

(E) Amorphous solids have a definite heat of fusion.

Choose the most appropriate answer from the options given below.

(A) (A), (B), (E) only

(B) (B), (D) only

(C) (C), (D) only

(D) (A), (C), (D) only

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(A) (A), (B), (E) only

(B) (B), (D) only

(C) (C), (D) only

(D) (A), (C), (D) only

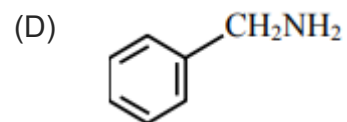
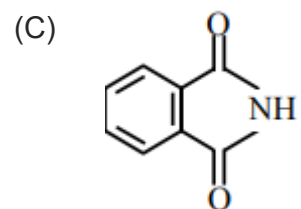
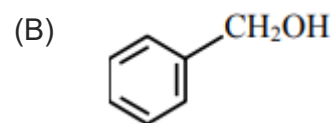
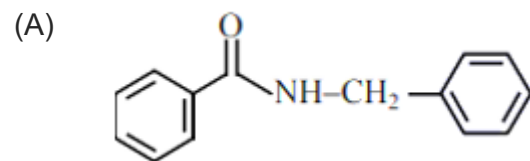
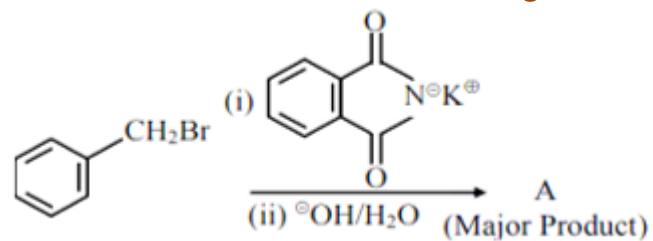
Solution:

Statements (A), (C) and (D) are correct statements. The other statements are corrected as follows –

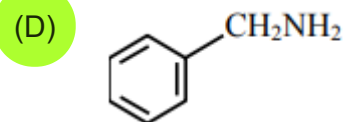
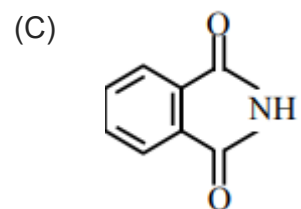
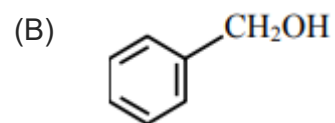
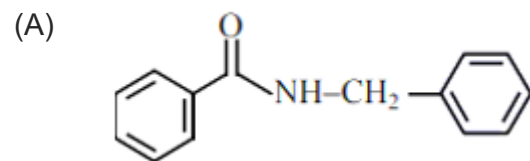
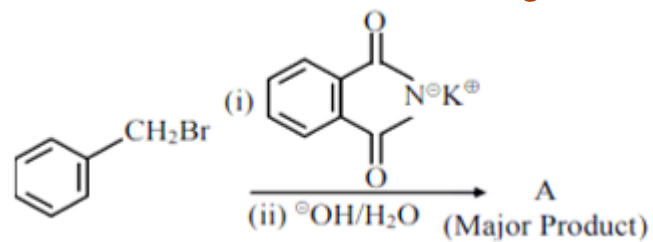
(B) Crystalline solids are anisotropic in nature.

(E) Amorphous solids do not have definite heat of fusion.

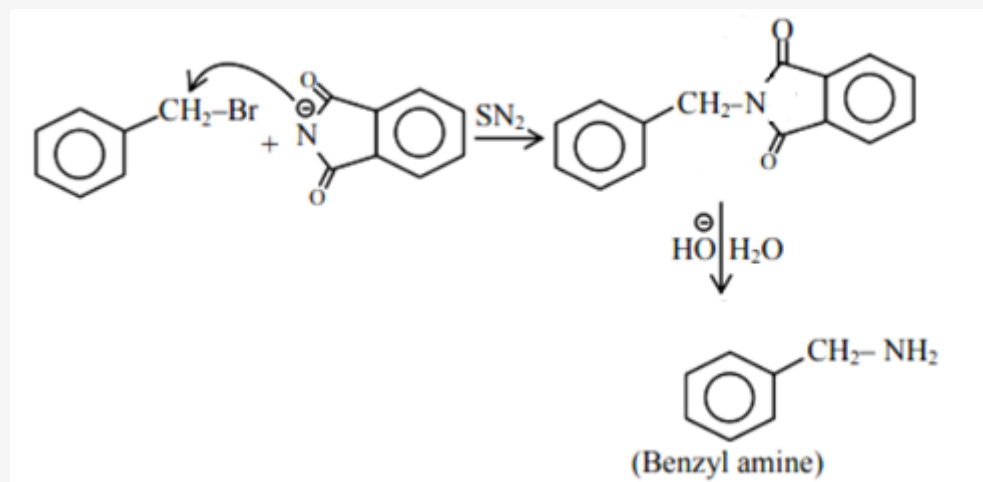
Q48: What is A in the following reaction?



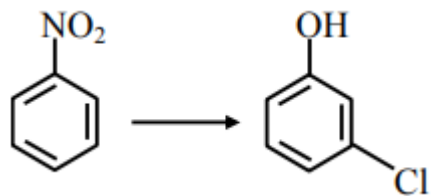
Q48: What is A in the following reaction?



Solution:

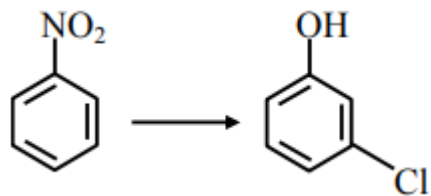


Q49: The correct sequence of correct reagents for the following transformation is :



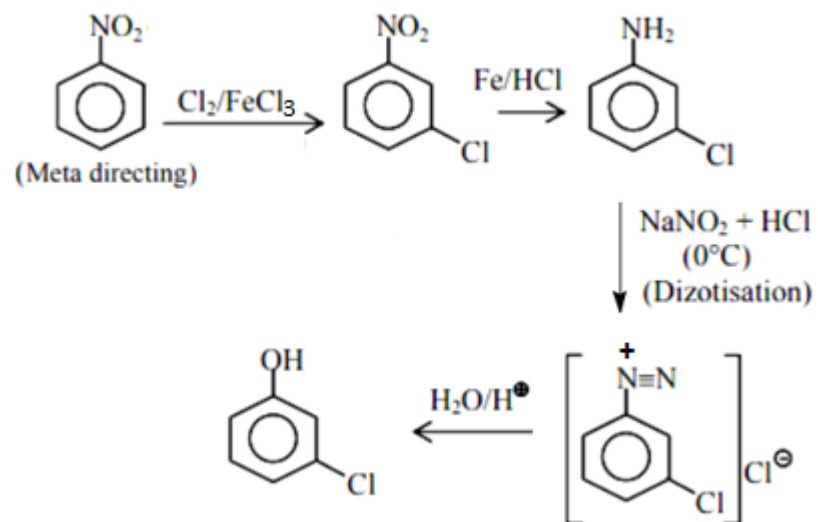
- (A) (i) Fe, HCl; (ii) Cl_2 , HCl; (iii) NaNO_2 , HCl, 0°C ; (iv) $\text{H}_2\text{O}/\text{H}^+$
- (B) (i) Fe, HCl; (ii) NaNO_2 , HCl, 0°C ; (iii) $\text{H}_2\text{O}/\text{H}^+$; (iv) Cl_2 , FeCl_3
- (C) (i) Cl_2 , FeCl_3 ; (ii) Fe, HCl; (iii) NaNO_2 , HCl, 0°C ; (iv) $\text{H}_2\text{O}/\text{H}^+$
- (D) (i) Cl_2 , FeCl_3 ; (ii) NaNO_2 , HCl, 0°C ; (iii) Fe, HCl; (iv) $\text{H}_2\text{O}/\text{H}^+$

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- (B) (i) Fe, HCl; (ii) NaNO_2 , HCl, 0°C ; (iii) $\text{H}_2\text{O}/\text{H}^+$; (iv) Cl_2 , FeCl_3
- (C) (i) Cl_2 , FeCl_3 ; (ii) Fe, HCl; (iii) NaNO_2 , HCl, 0°C ; (iv) $\text{H}_2\text{O}/\text{H}^+$
- (D) (i) Cl_2 , FeCl_3 ; (ii) NaNO_2 , HCl, 0°C ; (iii) Fe, HCl; (iv) $\text{H}_2\text{O}/\text{H}^+$

Solution:



Q50: The addition of silica during the extraction of copper from its sulphide ore

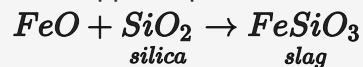
- (A) converts copper sulphide into copper silicate
- (B) converts iron oxide into iron silicate
- (C) reduces copper sulphide into metallic copper
- (D) reduces the melting point of the reaction mixture

Q50: The addition of silica during the extraction of copper from its sulphide ore

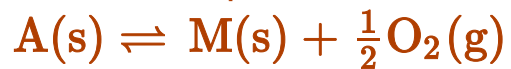
- (A) converts copper sulphide into copper silicate
- ☒ (B) converts iron oxide into iron silicate
- (C) reduces copper sulphide into metallic copper
- (D) reduces the melting point of the reaction mixture

Solution:

The copper ore is heated in a reverberatory furnace after mixing with silica. In the furnace, iron oxide 'slags off' as iron silicate and copper is produced in the form of copper matte.

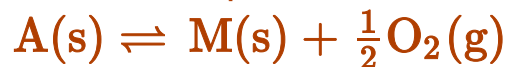


Q51: The equilibrium constant for the reaction



is $K_p = 4$. At equilibrium, the partial pressure of O_2 is _____ atm. (Round off to the nearest integer)

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Solution:

$$K_p = (P_{\text{O}_2})^{\frac{1}{2}} = 4$$
$$P_{\text{O}_2} = 16$$

Q52: When 400 mL of 0.2 M H_2SO_4 solution is mixed with 600 mL of 0.1 M NaOH solution, the increase in temperature of the final solution is _____ $\times 10^{-2}$ K. (Round off to the nearest integer).

[Use: $\text{H}^+ (\text{aq}) + \text{OH}^- (\text{aq}) \rightarrow \text{H}_2\text{O}$: $\Delta_r H = -57.1 \text{ kJ mol}^{-1}$]

Specific heat of $\text{H}_2\text{O} = 4.18 \text{ J K}^{-1} \text{ g}^{-1}$

Density of $\text{H}_2\text{O} = 1.0 \text{ g cm}^{-3}$

Assume no change in volume of solution on mixing.

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Specific heat of $\text{H}_2\text{O} = 4.18 \text{ J K}^{-1} \text{ g}^{-1}$

Density of $\text{H}_2\text{O} = 1.0 \text{ g cm}^{-3}$

Assume no change in volume of solution on mixing.

82.00

Solution:

mmoles of $\text{H}^+ = 2 \times 400 \times 0.2 = 160$ mmoles

mmoles of $\text{OH}^- = 600 \times 0.1 = 60$ mmoles

Thus, 60 mmoles of H^+ will react with 60 mmoles of OH^- .

Amount of heat liberated $= 60 \times 10^{-3} \times 57.1 \times 10^3 = 3426 \text{ J}$

Total volume of water $= 400 + 600 = 1000 \text{ mL}$

Thus, total mass of water $= 1000 \text{ g}$

Now, $q = ms\Delta T$

$$\Rightarrow \Delta T = \frac{q}{ms} = \frac{3426}{1000 \times 4.18} = 81.96 \times 10^{-2} \text{ K}$$



The above reaction is carried out in a vessel starting with partial pressure $P_{\text{SO}_2} = 250$ mbar, $P_{\text{O}_2} = 750$ mbar and $P_{\text{SO}_3} = 0$ bar. When the reaction is complete, the total pressure in the reaction vessel is _____ mbar. (Round off to the nearest integer).



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875.00

Solution:

	$2\text{SO}_2(\text{g})$	+	$\text{O}_2(\text{g})$	$\rightarrow$	$2\text{SO}_3(\text{g})$
Initial	250		750		0
Final	0 (limiting reagent)		625		250

Thus, total pressure = $625 + 250 = 875$ mbar.

Q54: 10.0 mL of 0.05 M KMnO_4 solution was consumed in a titration with 10.0 mL of given oxalic acid dihydrate solution. The strength of given oxalic acid solution is $\times 10^{-2}$ g/L.
(Round off to the nearest integer)

Q54: 10.0 mL of 0.05 M KMnO_4 solution was consumed in a titration with 10.0 mL of given oxalic acid dihydrate solution. The strength of given oxalic acid solution is $\times 10^{-2}$ g/L. (Round off to the nearest integer)

1575.00

Solution:

n-factor of $\text{KMnO}_4 = 5$

n-factor of $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O} = 2$

For complete titration, the number of equivalents must be equal for both substances.

$$n_1 M_1 V_1 = n_2 M_2 V_2$$

$$\Rightarrow 5 \times 0.05 \times 10 = 2 \times M_2 \times 10$$

$$\Rightarrow M_2 = \frac{5 \times 0.05}{2} = 0.125 \text{ M}$$

Now, strength = molarity $\times$ molar mass of solute

$$\text{Thus, strength} = 0.125 \times 126 = 15.75 \text{ g/L} = 1575 \times 10^{-2} \text{ g/L}$$

Q55: The total number of electrons in all bonding molecular orbitals of O_2^{2-} is

Q55: The total number of electrons in all bonding molecular orbitals of O_2^{2-} is

10.00

Solution:

O_2^{2-} (Total electrons = 18)

$$\text{EC} = (\sigma 1s)^2 (\sigma^* 1s)^2 (\sigma 2s)^2 (\sigma^* 2s)^2 (\sigma 2p_z)^2 (\pi 2p_x^2 = \pi 2p_y^2) (\pi^* 2p_x^2 = \pi^* 2p_y^2)$$

Total electrons in BMO = 10

Q56: 3 moles of metal complex with formula $\text{Co(en)}_2\text{Cl}_3$ gives 3 moles of silver chloride on treatment with excess of silver nitrate. The secondary valency of Co in the complex is _____.

Q56: 3 moles of metal complex with formula $\text{Co(en)}_2\text{Cl}_3$ gives 3 moles of silver chloride on treatment with excess of silver nitrate. The secondary valency of Co in the complex is _____.

6.00

Solution:

According to the given data, one mole of the complex will give one mole of silver chloride on treatment with excess of silver nitrate. This means the given complex has one chloride ion outside the coordination sphere and two chloride ions inside the coordination sphere. The formula can be written as $[\text{Co(en)}_2\text{Cl}_2] \text{Cl}$.

Thus, secondary valency = coordination number = 6.

Q57: In a solvent 50% of an acid HA dimerizes and the rest dissociates. The van't Hoff factor of the acid is _____ $\times 10^{-2}$. (Round off to the nearest integer)

Q57: In a solvent 50% of an acid HA dimerizes and the rest dissociates. The van't Hoff factor of the acid is _____ $\times 10^{-2}$. (Round off to the nearest integer)

125.00

Solution:

Let there was one mole of HA initially.

(i) Dissociation of 0.5 mole HA will produce 0.5 moles of H^+ and 0.5 moles of A^- .

(ii) Dimerisation of 0.5 mole HA will produce $\frac{0.5}{2}$ moles of $(HA)_2$

Thus, the total moles of particles after dissociation + association = $0.5 + 0.5 + \frac{0.5}{2} = 1.25$

Now, $i = \frac{\text{number of particles after dissociation + association}}{\text{number of particles before dissociation + association}}$

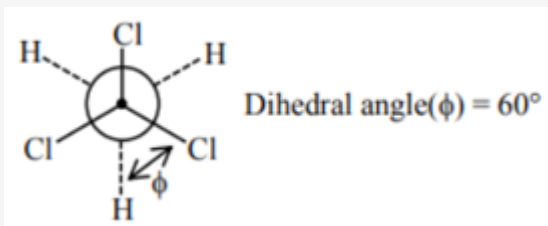
$$\Rightarrow i = \frac{1.25}{1} = 125 \times 10^{-2}$$

Q58: The dihedral angle in staggered form of Newman projection of 1, 1, 1-Trichloro ethane is degree. (Round off to the nearest integer)

Q58: The dihedral angle in staggered form of Newman projection of 1, 1, 1-Trichloro ethane is degree. (Round off to the nearest integer)

60.00

Solution:



Q59: For the first order reaction $A \rightarrow 2B$, 1 mole of reactant A gives 0.2 moles of B after 100 minutes. The half-life of the reaction is min. (Round off to the nearest integer).

[Use: $\ln 2 = 0.69$, $\ln 10 = 2.3$, $\ln 3 = 1.1$]

Q59: For the first order reaction $A \rightarrow 2B$, 1 mole of reactant A gives 0.2 moles of B after 100 minutes. The half-life of the reaction is min. (Round off to the nearest integer).

[Use: $\ln 2 = 0.69$, $\ln 10 = 2.3$, $\ln 3 = 1.1$]

690.00

Solution:

Given moles of A initially = 1.

After 100 mins, 0.2 moles of B are formed which means 0.1 moles of A are reacted.

Thus, moles of A remaining after 100 mins = 0.9.

Using first order integrated rate law, $\ln \frac{R_0}{R_t} = kt$

$$\Rightarrow \ln \frac{1}{0.9} = k \times 100$$

$$\Rightarrow \frac{\ln 10 - \ln 9}{100} = k$$

$$\text{Now, half life for first order reaction} = \frac{\ln 2}{k} = \left(\frac{\ln 2}{\ln 10 - 2 \ln 3} \right) \times 100 = \left(\frac{0.69}{2.3 - 2.2} \right) \times 100$$

Thus, half-life = 690 mins.

Q60: The cell



the cell potential $E_1 = 0.3095 \text{ V}$

For the cell



the cell potential = _____ $\times 10^{-2} \text{ V}$. (Round off the Nearest Integer).

[Use: $\frac{2.303 RT}{F} = 0.059$]

Q60: The cell



the cell potential $E_1 = 0.3095 \text{ V}$

For the cell



the cell potential = _____ $\times 10^{-2} \text{ V}$. (Round off the Nearest Integer).

[Use: $\frac{2.303 RT}{F} = 0.059$]

28.00

Solution:

The cell reaction is $\text{Cu} + 2\text{Ag}^+ \rightarrow 2\text{Ag} + \text{Cu}^{2+}$

Nernst equation can be written as - $E = E^\circ - \frac{0.059}{n} \log \frac{[\text{Cu}^{2+}]}{[\text{Ag}^+]^2}$

Substituting values for the first cell,

$$0.3095 = E^\circ - \frac{0.059}{2} \log \frac{10^{-1}}{10^{-4}}$$

$$\Rightarrow 0.3095 = E^\circ - \frac{0.059}{2} \times 3$$

$$\Rightarrow E^\circ = 0.3095 + 0.0885 = 0.398$$

For the second cell,

$$E_2 = 0.398 - \frac{0.059}{2} \log \frac{10^{-2}}{10^{-6}}$$

$$E_2 = 0.398 - \frac{0.059}{2} \times 4$$

$$E_2 = 0.398 - 0.118 = 0.28 \text{ V} = 28 \times 10^{-2} \text{ V}$$

Q61: Which of the following is the negation of the statement “for all $M > 0$, there exists $x \in S$ such that $x \geq M$ ”?

- (A) there exists $M > 0$, such that for all $x \geq M$ for all $x \in S$
- (B) there exists $M > 0$, there exists for $x \in S$ such that $x \geq M$
- (C) there exists $M > 0$, such that $x < M$ for all $x \in S$
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- (D) there exists $M > 0$, there exists $x \in S$ such that $x < M$

Solution:

P: for all $M > 0$, there exists $x \in S$ such that $x \geq M$.

$\sim P$: there exists $M > 0$, Such that $x < M$ for all $x \in S$

Negation of there exists is for all.

Q62: For real numbers α and $\beta \neq 0$, if the point of intersection of the straight lines $\frac{x-\alpha}{1} = \frac{y-1}{2} = \frac{z-1}{3}$ and $\frac{x-4}{\beta} = \frac{y-6}{3} = \frac{z-7}{3}$, lies on the plane $x + 2y - z = 8$, then $\alpha - \beta$ is equal to:

- (A) 5
- (B) 3
- (C) 7
- (D) 9

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(A) 5

(B) 3

(C) 7

(D) 9

Solution:

$$L_1 : \frac{x-\alpha}{1} = \frac{y-1}{2} = \frac{z-1}{3} = r_1$$

$$L_2 : \frac{x-4}{\beta} = \frac{y-6}{3} = \frac{z-7}{3} = r_2$$

$$L_1 (r_1 + \alpha, 2r_1 + 1, 3r_1 + 1)$$

$$L_2 (\beta r_2 + 4, 3r_2 + 6, 3r_2 + 7)$$

$$\therefore r_1 + \alpha = \beta r_2 + 4 \quad \dots (i)$$

$$2r_1 + 1 = 3r_2 + 6 \quad \dots (ii)$$

$$3r_1 + 1 = 3r_2 + 7 \quad \dots (iii)$$

From (ii) and (iii), we get

$$r_1 = 1, r_2 = -1$$

Putting the value of r_1 and r_2 in (1),

$$\alpha + \beta = 3 \quad \dots (iv)$$

$$L_1 (1 + \alpha, 3, 4)$$

Will satisfy the plane $x + 2y - z = 8$,

$$\Rightarrow \alpha = 5$$

$$\Rightarrow \beta = 3 - 5 = -2$$

$$\text{Now, } \alpha - \beta = 5 - (-2) = 7$$

Q63: The point P (a, b) undergoes the following three transformations successively:

(1) reflection about the line $y = x$.

(2) translation through 2 units along the positive direction of x-axis.

(3) rotation through angle $\frac{\pi}{4}$ about the origin in the anti-clockwise direction.

If the co-ordinates of the final position of the point P are $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$, then the value of $2a + b$ is equal to

(A) 9

(B) 5

(C) 13

(D) 7

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- (A) 9
- (B) 5
- (C) 13
- (D) 7

Solution:

Image of $A(a, b)$ along $y = x$ is $B(b, a)$. Translating it 2 units it becomes $C(b + 2, a)$.

Now, applying rotation theorem

$$-\frac{1}{\sqrt{2}} + \frac{7}{\sqrt{2}}i = ((b + 2) + ai) \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$\frac{-1}{\sqrt{2}} + \frac{7}{\sqrt{2}}i = \left(\frac{b+2}{\sqrt{2}} - \frac{a}{\sqrt{2}} \right) + i \left(\frac{b+2}{\sqrt{2}} + \frac{a}{\sqrt{2}} \right)$$

$$\Rightarrow b - a + 2 = -1 \quad \dots (1)$$

$$\text{and } b + 2 + a = 7 \quad \dots (2)$$

$$\Rightarrow a = 4; b = 1$$

$$\Rightarrow 2a + b = 9$$

Q64: Let $\alpha = \max_{x \in R} \{ 8^{2 \sin 3x} \cdot 4^{4 \cos 3x} \}$ and $\beta = \min_{x \in R} \{ 8^{2 \sin 3x} \cdot 4^{4 \cos 3x} \}$

If $8x^2 + bx + c = 0$ is a quadratic equation whose roots are $\alpha^{\frac{1}{5}}$ and $\beta^{\frac{1}{5}}$, then the value of $c - b$ is equal to:

(A) 43

(B) 42

(C) 50

(D) 47

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(A) 43

(B) 42

(C) 50

(D) 47

Solution:

$$\alpha = \max \{8^{2 \sin 3x} \cdot 4^{4 \cos 3x}\} = \max \{2^{6 \sin 3x + 8 \cos 3x}\}$$

$$\text{and } \beta = \min \{8^{2 \sin 3x} \cdot 4^{4 \cos 3x}\} = \min \{2^{6 \sin 3x + 8 \cos 3x}\}$$

Now range of $6 \sin 3x + 8 \cos 3x$

$$= \left[-\sqrt{6^2 + 8^2}, +\sqrt{6^2 + 8^2} \right] = [-10, 10]$$

$$\alpha = 2^{10} \text{ \& } \beta = 2^{-10}$$

$$\text{So, } \alpha^{\frac{1}{5}} = 2^2 = 4$$

$$\Rightarrow \beta^{\frac{1}{5}} = 2^{-2} = \frac{1}{4}$$

quadratic $8x^2 + bx + c = 0$,

$$c - b = 8 \times [(\text{product of roots}) + (\text{sum of roots})]$$

$$= 8 \times \left[4 \times \frac{1}{4} + 4 + \frac{1}{4} \right] = 8 \times \left[\frac{21}{4} \right] = 42$$

Q65: Let $f : R \rightarrow R$ be defined as

$f(x + y) + f(x - y) = 2f(x)f(y)$, $f\left(\frac{1}{2}\right) = -1$. Then, the value of

$$\sum_{k=1}^{20} \frac{1}{\sin(k) \sin(k+f(k))}$$

is equal to:

- (A) $\operatorname{cosec}^2(1) \operatorname{cosec}(21) \sin(20)$
- (B) $\sec^2(1) \sec(21) \cos(20)$
- (C) $\operatorname{cosec}^2(21) \cos(20) \cos(2)$
- (D) $\sec^2(21) \sin(20) \sin(2)$

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(C) $\operatorname{cosec}^2(21) \cos(20) \cos(2)$

(D) $\sec^2(21) \sin(20) \sin(2)$

Solution:

$$f(x) = \cos ax$$

$$\because f\left(\frac{1}{2}\right) = -1$$

$$\text{So, } -1 = \cos \frac{a}{2}$$

$$\Rightarrow a = 2(2n+1)\pi$$

$$\text{Thus } f(x) = \cos 2(2n+1)\pi x$$

Now k is natural number

$$\text{Thus } f(k) = 1$$

$$\sum_{k=1}^{20} \frac{1}{\sin k \sin(k+1)} = \frac{1}{\sin 1} \sum_{k=1}^{20} \left[\frac{\sin((k+1)-k)}{\sin k \sin(k+1)} \right]$$

$$= \frac{1}{\sin 1} \sum_{k=1}^{20} (\cot k - \cot(k+1))$$

$$= \frac{\cot 1 - \cot 21}{\sin 1} = \operatorname{cosec}^2 1 \operatorname{cosec}(21) \cdot \sin 20$$

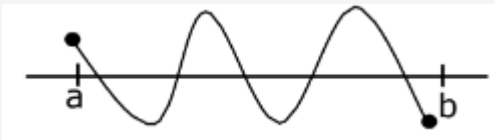
Q66: Let $f : (a, b) \rightarrow \mathbb{R}$ be twice differentiable function such that $f(x) = \int_a^x g(t) dt$ for a differentiable function $g(x)$. If $f(x) = 0$ has exactly five distinct roots in (a, b) , then $g(x) g'(x) = 0$ has at least:

- (A) seven roots in (a, b)
- (B) five roots in (a, b)
- (C) three roots in (a, b)
- (D) twelve roots in (a, b)

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- (A) seven roots in (a, b)
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- (C) three roots in (a, b)
- (D) twelve roots in (a, b)

Solution:



$$f(x) = \int_a^x g(t) dt$$

$$f(x) \rightarrow 5$$

$$f'(x) \rightarrow 4$$

$$g(x) \rightarrow 4$$

$$g'(x) \rightarrow 3$$

Q67: Let C be the set of all complex numbers. Let

$$S_1 = \{z \in C : |z - 2| \leq 1\} \text{ and } S_2 = \{z \in C : z(1 + i) + \bar{z}(1 - i) \geq 4\}$$

Then, the maximum value of $\left|z - \frac{5}{2}\right|^2$ for $z \in S_1 \cap S_2$ is equal to:

(A) $\frac{3+2\sqrt{2}}{4}$

(B) $\frac{5+2\sqrt{2}}{2}$

(C) $\frac{3+2\sqrt{2}}{2}$

(D) $\frac{5+2\sqrt{2}}{4}$

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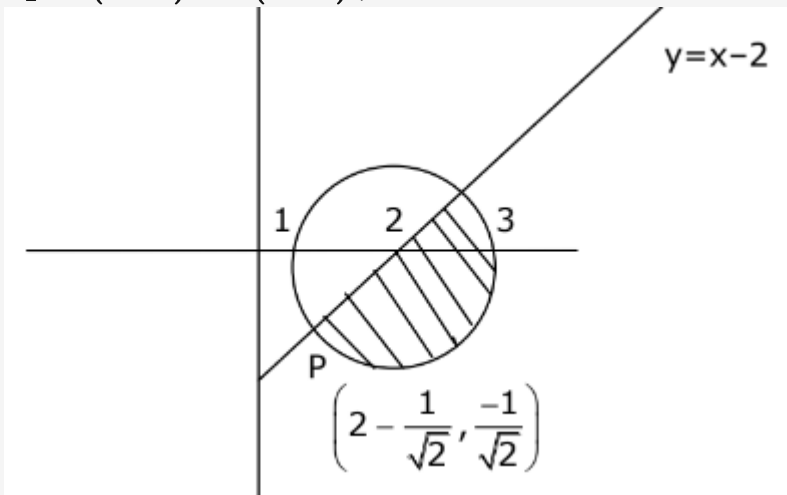
(C) $\frac{3+2\sqrt{2}}{2}$

(D) $\frac{5+2\sqrt{2}}{4}$

Solution:

$$S_1 : |z - 2| \leq 1$$

$$S_2 : z(1 + i) + \bar{z}(1 - i) \geq 4$$



$$z = x + iy$$

$$(x + iy)(1 + i) + (x - iy)(1 - i) \geq 4$$

$$x + i(x + y) - y + x - i(x + y) - y \geq 4$$

$$2x - 2y \geq 4$$

$$\Rightarrow y \leq x - 2$$

$$\text{For } P(x - 2)^2 + (x - 2)^2 = 1$$

$$\Rightarrow x = 2 - \frac{1}{\sqrt{2}}, 2 + \frac{1}{\sqrt{2}}$$

$$\therefore \text{from diagram x-coordinate of } P = 2 - \frac{1}{\sqrt{2}}$$

$$\therefore P = \left(2 - \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$$

$$\begin{aligned}\therefore \left| z - \frac{5}{2} \right|_{\max}^2 &= \left| 2 - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i - \frac{5}{2} \right|^2 \\ &= \left| \left(\frac{-1}{2} - \frac{1}{\sqrt{2}} \right) - \frac{i}{\sqrt{2}} \right|^2 = \frac{5+2\sqrt{2}}{4}\end{aligned}$$

Q68: Two sides of a parallelogram are along the lines $4x + 5y = 0$ and $7x + 2y = 0$. If the equation of one of the diagonals of the parallelogram is $11x + 7y = 9$, then other diagonal passes through the point:

- (A) $(1, 3)$
- (B) $(1, 2)$
- (C) $(2, 2)$
- (D) $(2, 1)$

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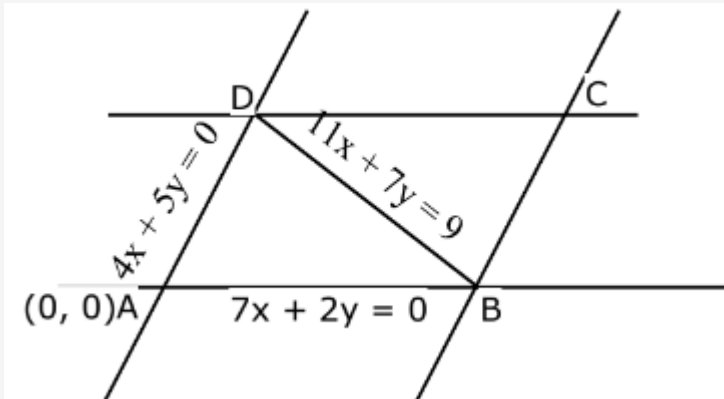
(B) $(1, 2)$

(C) $(2, 2)$

(D) $(2, 1)$

Solution:

Both the lines pass through origin.



point D is equal of intersection of $4x + 5y = 0$ & $11x + 7y = 9$

So, coordinates of point $D = \left(\frac{5}{3}, -\frac{4}{3}\right)$

Also, point B is point of intersection of $7x + 2y = 0$ & $11x + 7y = 9$

So, coordinates of point $B = \left(-\frac{2}{3}, \frac{7}{3}\right)$

diagonals of parallelogram intersect at middle let middle point of B,D

$$\Rightarrow \left(\frac{\frac{5}{3} - \frac{2}{3}}{2}, \frac{-\frac{4}{3} + \frac{7}{3}}{2} \right) = \left(\frac{1}{2}, \frac{1}{2} \right)$$

equation of diagonal AC

$$\Rightarrow (y - 0) = \frac{\frac{1}{2} - 0}{\frac{1}{2} - 0} (x - 0)$$

$$y = x$$

Diagonal AC passes through (2, 2).

Q69: Let $f : [0, \infty) \rightarrow [0, 3]$ be a function defined by

$$f(x) = \begin{cases} \max \{ \sin t : 0 \leq t \leq x \}, & 0 \leq x \leq \pi \\ 2 + \cos x, & x > \pi \end{cases}$$

Then which of the following is true?

- (A) f is differentiable everywhere in $(0, \infty)$
- (B) f is continuous everywhere but not differentiable exactly at two points in $(0, \infty)$
- (C) f is not continuous exactly at two points in $(0, \infty)$
- (D) f is continuous everywhere but not differentiable exactly at one point in $(0, \infty)$

Q69: Let $f : [0, \infty) \rightarrow [0, 3]$ be a function defined by

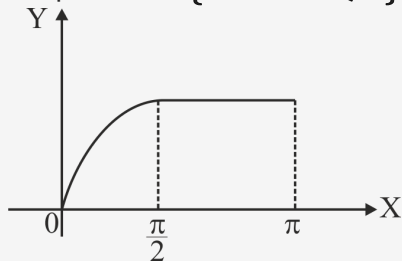
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Then which of the following is true?

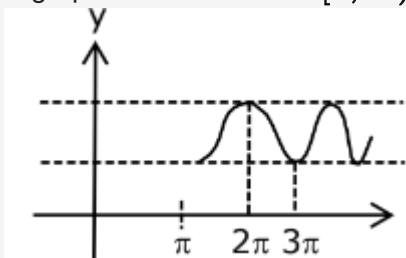
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- (C) f is not continuous exactly at two points in $(0, \infty)$
- (D) f is continuous everywhere but not differentiable exactly at one point in $(0, \infty)$

Solution:

Graph of $\max \{\sin t : 0 \leq t \leq x\}$ in $x \in [0, \pi]$

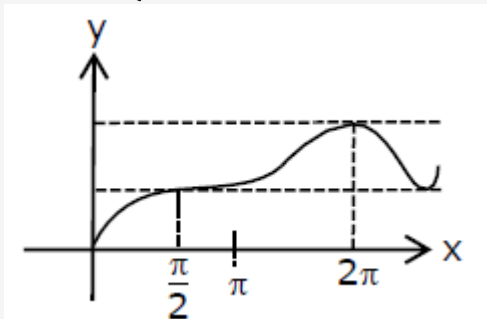


& graph of $\cos x$ for $x \in [\pi, \infty)$



So graph of

$$f(x) = \begin{cases} \max\{\sin t : 0 \leq t \leq x, & 0 \leq x \leq \pi \\ 2 + \cos x & x > \pi \end{cases}$$



$f(x)$ is differentiable everywhere in $(0, \infty)$

Q70: The area of the region bounded by $y - x = 2$ and $x^2 = y$ is equal to

(A) $\frac{4}{3}$

(B) $\frac{9}{2}$

(C) $\frac{16}{3}$

(D) $\frac{2}{3}$

Q70: The area of the region bounded by $y - x = 2$ and $x^2 = y$ is equal to

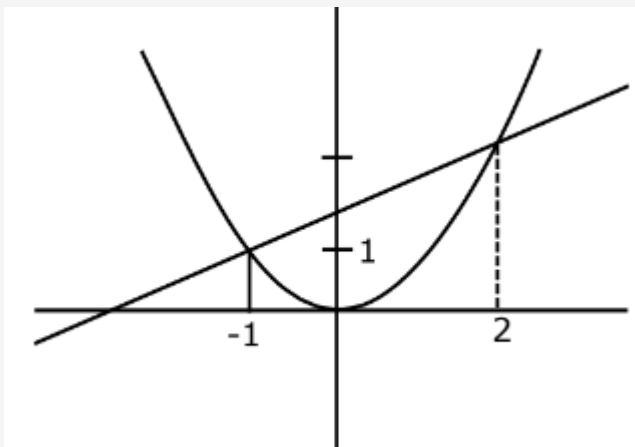
(A) $\frac{4}{3}$

(B) $\frac{9}{2}$

(C) $\frac{16}{3}$

(D) $\frac{2}{3}$

Solution:



$$y - x = 2, x^2 = y$$

$$\text{Now, } x^2 = 2 + x$$

$$\Rightarrow x^2 - x - 2 = 0$$

$$\Rightarrow (x + 1)(x - 2) = 0$$

$$\text{Area} = \int_{-1}^2 (2 + x - x^2) dx$$

$$= \left| 2x + \frac{x^2}{2} - \frac{x^3}{3} \right|_{-1}^2$$

$$= \left(4 + 2 - \frac{8}{3} \right) - \left(-2 + \frac{1}{2} + \frac{1}{3} \right)$$

$$= 6 - 3 + 2 - \frac{1}{2} = \frac{9}{2}$$

Q71: Let the mean and variance of the frequency distribution

$x :$	$x_1 = 2$	$x_2 = 6$	$x_3 = 8$	$x_4 = 9$
$f :$	4	4	α	β

be 6 and 6.8 respectively. If x_3 is changed from 8 to 7, then the mean for the new data will be:

- (A) $\frac{16}{3}$
- (B) 4
- (C) $\frac{17}{3}$
- (D) 5

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be 6 and 6.8 respectively. If x_3 is changed from 8 to 7, then the mean for the new data will be:

(A) $\frac{16}{3}$

(B) 4

(C) $\frac{17}{3}$

(D) 5

Solution:

$$\text{Given } 32 + 8\alpha + 9\beta = (8 + \alpha + \beta) \times 6$$

$$\Rightarrow 2\alpha + 3\beta = 16 \dots (i)$$

$$\text{Also, } 4 \times 16 + 4 \times \alpha + 9\beta = (8 + \alpha + \beta) \times 6.8$$

$$\Rightarrow 640 + 40\alpha + 90\beta = 544 + 68\alpha + 68\beta$$

$$\Rightarrow 28\alpha - 22\beta = 96$$

$$\Rightarrow 14\alpha - 11\beta = 48 \dots (ii)$$

$$\text{from (i) \& (ii) } \alpha = 5 \text{ \& } \beta = 2$$

$$\text{so, new mean} = \frac{32+35+18}{15} = \frac{85}{15} = \frac{17}{3}$$

Q72: A possible value of ' x ', for which the ninth term in the expansion of

$\left\{ 3^{\log_3 \sqrt{25^{x-1}+7}} + 3^{\left(-\frac{1}{8}\right)\log_3 (5^{x-1}+1)} \right\}^{10}$ in the increasing powers of $3^{\left(\frac{1}{8}\right)\log_3 (5^{x-1}+1)}$ is

equal to 180, is:

- (A) 2
- (B) 1
- (C) 0
- (D) -1

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$\left\{ 3^{\log_3 \sqrt{25^{x-1}+7}} + 3^{\left(-\frac{1}{8}\right)\log_3 (5^{x-1}+1)} \right\}^{10}$ in the increasing powers of $3^{\left(\frac{1}{8}\right)\log_3 (5^{x-1}+1)}$ is

equal to 180, is:

(A) 2

(B) 1

(C) 0

(D) -1

Solution:

$${}^{10}C_8 \left(25^{(x-1)} + 7 \right) \times \left(5^{(x-1)} + 1 \right)^{-1} = 180$$

$$\Rightarrow \frac{25^{x-1}+7}{5^{(x-1)}+1} = 4$$

$$\Rightarrow \frac{t^2+7}{t+1} = 4;$$

$$\Rightarrow t = 1, 3 = 5^{x-1}$$

$$\Rightarrow x - 1 = 0 \text{ (one of the possible value).}$$

$$\Rightarrow x = 1$$

Q73: Let $y = y(x)$ be the solution of the differential equation

$(x - x^3) dy = (y + yx^2 - 3x^4) dx, x > 2$. If $y(3) = 3$, then $y(4)$ is equal to:

- (A) 8
- (B) 12
- (C) 16
- (D) 4

Q73: Let $y = y(x)$ be the solution of the differential equation $(x - x^3) dy = (y + yx^2 - 3x^4) dx, x > 2$. If $y(3) = 3$, then $y(4)$ is equal to:

- (A) 8
- (B) 12
- (C) 16
- (D) 4

Solution:

$$\begin{aligned}(x - x^3) dy &= (y + yx^2 - 3x^4) dx \\ \Rightarrow xdy - ydx &= (yx^2 - 3x^4) dx + x^3 dy \\ \Rightarrow \frac{xdy - ydx}{x^2} &= (ydx + xdy) - 3x^2 dx \\ \Rightarrow d\left(\frac{y}{x}\right) &= d(xy) - d(x^3)\end{aligned}$$

Integrate

$$\frac{y}{x} = xy - x^3 + c$$

$$y(3) = 3$$

$$\Rightarrow \frac{3}{3} = 3 \times 3 - 3^3 + c$$

$$\Rightarrow c = 19$$

$$\therefore \frac{y}{x} = xy - x^3 + 19$$

$$\text{at } x = 4, \frac{y}{4} = 4y - 64 + 19$$

$$15y = 4 \times 45$$

$$\Rightarrow y = 12$$

Q74: Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a} = \vec{b} \times (\vec{b} \times \vec{c})$. If magnitudes of the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are $\sqrt{2}$, 1 and 2 respectively and the angle between $\vec{b}$ and $\vec{c}$ is θ ($0 < \theta < \frac{\pi}{2}$), then the value of $1 + \tan \theta$ is equal to:

(A) $\frac{\sqrt{3}+1}{\sqrt{3}}$

(B) 2

(C) $\sqrt{3} + 1$

(D) 1

Q74: Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a} = \vec{b} \times (\vec{b} \times \vec{c})$. If magnitudes of the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are $\sqrt{2}$, 1 and 2 respectively and the angle between $\vec{b}$ and $\vec{c}$ is θ ($0 < \theta < \frac{\pi}{2}$), then the value of $1 + \tan \theta$ is equal to:

(A) $\frac{\sqrt{3}+1}{\sqrt{3}}$

(B) 2

(C) $\sqrt{3} + 1$

(D) 1

Solution:

$$\vec{a} = \left(\vec{b} \cdot \vec{c} \right) \vec{b} - \left(\vec{b} \cdot \vec{b} \right) \vec{c}$$

$$= 1.2 \cos \theta \vec{b} - \vec{c}$$

$$\Rightarrow \vec{a} = 2 \cos \theta \vec{b} - \vec{c}$$

$$\left| \vec{a} \right|^2 = (2 \cos \theta)^2 + 2^2 - 2.2 \cos \theta \vec{b} \cdot \vec{c}$$

$$\Rightarrow 2 = 4 \cos^2 \theta + 4 - 4 \cos \theta . 2 \cos \theta$$

$$\Rightarrow -2 = -4 \cos^2 \theta$$

$$\Rightarrow \cos^2 \theta = \frac{1}{2}$$

$$\Rightarrow \sec^2 \theta = 2$$

$$\Rightarrow \tan^2 \theta = 1$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

$$1 + \tan \theta = 2.$$

Q75: A student appeared in an examination consisting of 8 true – false type questions. The student guesses the answers with equal probability. The smallest value of n , so that the probability of guessing at least ' n ' correct answers is less than $\frac{1}{2}$, is:

- (A) 5
- (B) 3
- (C) 6
- (D) 4

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(A) 5

(B) 3

(C) 6

(D) 4

Solution:

$$P(E) < \frac{1}{2}$$

$$\Rightarrow \sum_{r=n}^8 {}^8C_r \left(\frac{1}{2}\right)^{8-r} \left(\frac{1}{2}\right)^r < \frac{1}{2}$$

$$\Rightarrow \sum_{r=n}^8 {}^8C_r \left(\frac{1}{2}\right)^8 < \frac{1}{2}$$

$$\Rightarrow {}^8C_n + {}^8C_{n+1} + \dots + {}^8C_8 < 128$$

$$\Rightarrow 256 - ({}^8C_0 + {}^8C_1 + \dots + {}^8C_{n-1}) < 128$$

$$\Rightarrow {}^8C_0 + {}^8C_1 + \dots + {}^8C_{n-1} > 128$$

$$\Rightarrow n - 1 \geq 4$$

$$\Rightarrow n \geq 5$$

Q76: If $\tan\left(\frac{\pi}{9}\right), x, \tan\left(\frac{7\pi}{18}\right)$ are in arithmetic progression and $\tan\left(\frac{\pi}{9}\right), y, \tan\left(\frac{5\pi}{18}\right)$ are also in arithmetic progression, then $|x - 2y|$ is equal to:

- (A) 0
- (B) 3
- (C) 4
- (D) 1

Q76: If $\tan\left(\frac{\pi}{9}\right), x, \tan\left(\frac{7\pi}{18}\right)$ are in arithmetic progression and $\tan\left(\frac{\pi}{9}\right), y, \tan\left(\frac{5\pi}{18}\right)$ are also in arithmetic progression, then $|x - 2y|$ is equal to:

(A) 0

(B) 3

(C) 4

(D) 1

Solution:

$$x = \frac{1}{2} \left(\tan \frac{\pi}{9} + \tan \frac{7\pi}{18} \right)$$

$$\text{and } 2y = \tan \frac{\pi}{9} + \tan \frac{5\pi}{18}$$

$$\text{so, } x - 2y = \frac{1}{2} \left(\tan \frac{\pi}{9} + \tan \frac{7\pi}{18} \right) - \left(\tan \frac{\pi}{9} + \tan \frac{5\pi}{18} \right)$$

$$\Rightarrow |x - 2y| = \left| \frac{\cot \frac{\pi}{9} - \tan \frac{\pi}{9}}{2} - \tan \frac{5\pi}{18} \right|$$

$$= \left| \cot \frac{2\pi}{9} - \cot \frac{2\pi}{9} \right| = 0$$

$$(\text{as } \tan \frac{5\pi}{18} = \cot \frac{2\pi}{9}; \tan \frac{7\pi}{18} = \cot \frac{\pi}{9})$$

Q77: Let N be the set of natural numbers and a relation R on N be defined by $R = \{(x, y) \in N \times N : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}$. Then the relation R is:

- (A) reflexive and symmetric, but not transitive
- (B) reflexive but neither symmetric nor transitive
- (C) an equivalence relation
- (D) symmetric but neither reflexive nor transitive

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- (A) reflexive and symmetric, but not transitive
- (B) reflexive but neither symmetric nor transitive
- (C) an equivalence relation
- (D) symmetric but neither reflexive nor transitive

Solution:

$$\begin{aligned}x^3 - 3x^2y - xy^2 + 3y^3 &= 0 \\ \Rightarrow x(x - y)(x + y) - 3y(x - y)(x + y) &= 0 \\ \Rightarrow (x - 3y)(x - y)(x + y) &= 0\end{aligned}$$

Now, $x = y \forall (x, y) \in N \times N$ so reflexive

But not symmetric & transitive

See, (3,1) satisfies but (1,3) does not.

Also (9,3) & (3,1) satisfies but (9,1) does not

Q78: The value of $\lim_{x \rightarrow 0} \left(\frac{x}{\sqrt[8]{1-\sin x} - \sqrt[8]{1+\sin x}} \right)$ is equal to:

(A) -1

(B) -4

(C) 0

(D) 4

Q78: The value of $\lim_{x \rightarrow 0} \left(\frac{x}{\sqrt[8]{1-\sin x} - \sqrt[8]{1+\sin x}} \right)$ is equal to:

(A) -1

(B) -4

(C) 0

(D) 4

Solution:

Rationalize denominator three times

$$\lim_{x \rightarrow 0} \frac{x \left\{ (1-\sin x)^{\frac{1}{8}} + (1+\sin x)^{\frac{1}{8}} \right\} \left\{ (1-\sin x)^{\frac{1}{4}} + (1+\sin x)^{\frac{1}{4}} \right\} \left\{ (1-\sin x)^{\frac{1}{2}} + (1+\sin x)^{\frac{1}{2}} \right\}}{(1-\sin x - 1 - \sin x)}$$

$$\lim_{x \rightarrow 0} \frac{8x}{-2 \sin x} = -4$$

Q79: Consider a circle C which touches the y-axis at (0, 6) and cuts off an intercept $6\sqrt{5}$ on the x -axis. Then the radius of the circle C is equal to:

(A) $\sqrt{82}$

(B) 9

(C) 8

(D) $\sqrt{53}$

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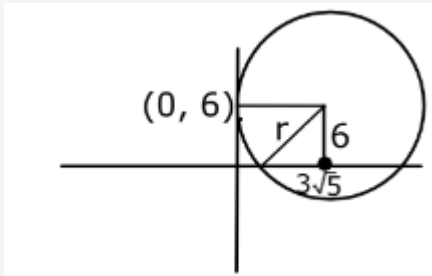
(A) $\sqrt{82}$

(B) 9

(C) 8

(D) $\sqrt{53}$

Solution:



$$\begin{aligned} r &= \sqrt{6^2 + (3\sqrt{5})^2} \\ &= \sqrt{36 + 45} = 9 \end{aligned}$$

Q80: Let A and B be two 3×3 real matrices such that $(A^2 - B^2)$ is invertible matrix. If $A^5 = B^5$ and $A^3 B^2 = A^2 B^3$, then the value of the determinant of the matrix $A^3 + B^3$ is equal to :

- (A) 0
- (B) 2
- (C) 1
- (D) 4

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(A) 0

(B) 2

(C) 1

(D) 4

Solution:

$$C = A^2 - B^2; |C| \neq 0$$

$$A^5 = B^5 \text{ and } A^3 B^2 = A^2 B^3$$

$$\text{Now, } A^5 - A^3 B^2 = B^5 - A^2 B^3$$

$$\Rightarrow A^3 (A^2 - B^2) + B^3 (A^2 - B^2) = 0$$

$$\Rightarrow (A^3 + B^3) (A^2 - B^2) = 0$$

Since $A^2 - B^2$ is non-singular

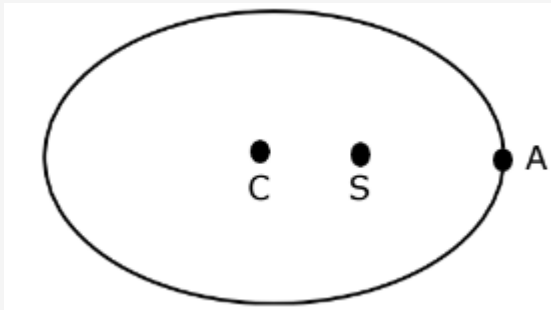
$$A^3 + B^3 = 0$$

Q81: Let E be an ellipse whose axes are parallel to the co-ordinates axes, having its centre at $(3, -4)$, one focus at $(4, -4)$ and one vertex at $(5, -4)$. If $mx - y = 4, m > 0$ is a tangent to the ellipse E , then the value of $5m^2$ is equal to

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3.00

Solution:



And $A(5, -4)$

Hence, $a = 2$ & $ae = 1$

$$\Rightarrow e = \frac{1}{2}$$

$$\Rightarrow b^2 = 3.$$

$$\text{So, } E: \frac{(x-3)^2}{4} + \frac{(y+4)^2}{3} = 1$$

Intersecting with given tangent.

$$\frac{x^2 - 6x + 9}{4} + \frac{m^2 x^2}{3} = 1$$

Now, $D = 0$ (as it is tangent)

$$\text{SO, } 5m^2 = 3.$$

Q82: If the real part of the complex number $z = \frac{3+2i \cos \theta}{1-3i \cos \theta}$, $\theta \in \left(0, \frac{\pi}{2}\right)$ is zero, then the value of $\sin^2 3\theta + \cos^2 \theta$ is equal to _____.

Q82: If the real part of the complex number $z = \frac{3+2i \cos \theta}{1-3i \cos \theta}$, $\theta \in \left(0, \frac{\pi}{2}\right)$ is zero, then the value of $\sin^2 3\theta + \cos^2 \theta$ is equal to _____.

1.00

Solution:

$$\begin{aligned} z &= \frac{3+2i \cos \theta}{1-3i \cos \theta} \\ &= \frac{(3+2i \cos \theta)(1+3i \cos \theta)}{(1-3i \cos \theta)(1+3i \cos \theta)} \\ &= \frac{(3-6\cos^2 \theta) + (11i \cos \theta)}{(1+9\cos^2 \theta)} \end{aligned}$$

$$\operatorname{Re}(z) = 0$$

$$\Rightarrow 3 - 6\cos^2 \theta = 0$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

$$\text{Hence, } \sin^2 3\theta + \cos^2 \theta = 1.$$

Q83: If $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ and $M = A + A^2 + A^3 + \dots + A^{20}$, then the sum of the all the elements of the matrix M is equal to _____.

Q83: If $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ and $M = A + A^2 + A^3 + \dots + A^{20}$, then the sum of the all the elements of the matrix M is equal to _____.

2020.00

Solution:

$$A^n = \begin{bmatrix} 1 & n & \frac{n^2+n}{2} \\ 0 & 1 & n \\ 0 & 0 & 1 \end{bmatrix}$$

So, required sum

$$\begin{aligned} &= 20 \times 3 + 2 \times \left(\frac{20 \times 21}{2} \right) + \sum_{r=1}^{20} \left(\frac{r^2+r}{2} \right) \\ &= 60 + 420 + 105 + 35 \times 41 = 2020 \end{aligned}$$

Q84: The distance of the point $P(3, 4, 4)$ from the point of intersection of the line joining the points $Q(3, -4, -5)$ and $R(2, -3, 1)$ and the plane $2x + y + z = 7$, is equal to _____.

Q84: The distance of the point $P(3, 4, 4)$ from the point of intersection of the line joining the points $Q(3, -4, -5)$ and $R(2, -3, 1)$ and the plane $2x + y + z = 7$, is equal to _____.

7.00

Solution:

$$\overrightarrow{QR} : \frac{x-3}{1} = \frac{y+4}{-1} = \frac{z+5}{-6} = r$$

$$\Rightarrow (x, y, z) \equiv (r+3, -r-4, -6r-5)$$

Now, satisfying it in the given plane.

$$2(r+3) + (-r-4) + (-6r-5) = 7$$

We get $r = -2$.

so, required point of intersection is $T(1, -2, 7)$

Hence, $PT = 7$.

Q85: The number of real roots of the equation $e^{4x} - e^{3x} - 4e^{2x} - e^x + 1 = 0$ is equal to

_____.

Q85: The number of real roots of the equation $e^{4x} - e^{3x} - 4e^{2x} - e^x + 1 = 0$ is equal to

2.00

Solution:

$$t^4 - t^3 - 4t^2 - t + 1 = 0, e^x = t > 0$$

$$\Rightarrow t^2 - t - 4 - \frac{1}{t} + \frac{1}{t^2} = 0$$

$$\Rightarrow \alpha^2 - \alpha - 6 = 0, \alpha = t + \frac{1}{t} \geq 2$$

$$\Rightarrow \alpha = 3, -2 \text{ (reject)}$$

$$\Rightarrow t + \frac{1}{t} = 3$$

$$\Rightarrow \text{The number of real roots} = 2$$

Q86: Let n be a non-negative integer. Then the number of divisors of the form " $4n + 1$ " of the number $(10)^{10} \cdot (11)^{11} \cdot (13)^{13}$ is equal to _____.

Q86: Let n be a non-negative integer. Then the number of divisors of the form " $4n + 1$ " of the number $(10)^{10} \cdot (11)^{11} \cdot (13)^{13}$ is equal to _____.

924.00

Solution:

$$N = 2^{10} \times 5^{10} \times 11^{11} \times 13^{13}$$

Now, power of 2 must be zero,

power of 5 can be anything,

power of 13 can be anything. But,

power of 11 should be even.

So, required number of divisors is

$$1 \times 11 \times 14 \times 6 = 924$$

Q87: If $\int_0^\pi (\sin^3 x) e^{-\sin^2 x} dx = \alpha - \frac{\beta}{e} \int_0^1 \sqrt{t} e^t dt$, then $\alpha + \beta$ is equal to _____.

Q87: If $\int_0^\pi (\sin^3 x) e^{-\sin^2 x} dx = \alpha - \frac{\beta}{e} \int_0^1 \sqrt{t} e^t dt$, then $\alpha + \beta$ is equal to _____.

5.00

Solution:

$$\begin{aligned}
 I &= 2 \int_0^{\frac{\pi}{2}} \sin^3 x e^{-\sin^2 x} dx \\
 &= 2 \int_0^{\frac{\pi}{2}} \sin x e^{-\sin^2 x} dx + \underbrace{\int_0^{\frac{\pi}{2}} \cos x e^{-\sin^2 x}}_{\text{I}} \underbrace{(-\sin 2x) dx}_{\text{II}} \\
 &= 2 \int_0^{\frac{\pi}{2}} \sin x e^{-\sin^2 x} dx + \left[\cos x e^{-\sin^2 x} \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \sin x e^{-\sin^2 x} dx \\
 &= 3 \int_0^{\frac{\pi}{2}} \sin x e^{-\sin^2 x} dx - 1 \\
 &= \frac{3}{2} \int_{-1}^0 \frac{e^\alpha d\alpha}{\sqrt{1+\alpha}} - 1 \text{ (Put } -\sin^2 x = \alpha) \\
 &= \frac{3}{2} \int_{-1}^0 \frac{e^\alpha d\alpha}{\sqrt{1+\alpha}} - 1 \text{ (Put } 1 + \alpha = x) \\
 &= \frac{3}{2e} \int_{-1}^0 \underbrace{e^x}_{\text{I}} \underbrace{\frac{1}{\sqrt{x}}}_{\text{II}} dx - 1 \\
 &= 2 - \frac{3}{e} \int_0^1 e^x \sqrt{x} dx
 \end{aligned}$$

Hence, $\alpha + \beta = 5$

Q88: Let $\vec{a} = \hat{i} - \alpha\hat{j} + \beta\hat{k}$, $\vec{b} = 3\hat{i} + \beta\hat{j} - \alpha\hat{k}$ and $\vec{c} = -\alpha\hat{i} - 2\hat{j} + \hat{k}$, where α and β are integers. If $\vec{a} \cdot \vec{b} = -1$ and $\vec{b} \cdot \vec{c} = 10$, then $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is equal to _____.

Q88: Let $\vec{a} = \hat{i} - \alpha\hat{j} + \beta\hat{k}$, $\vec{b} = 3\hat{i} + \beta\hat{j} - \alpha\hat{k}$ and $\vec{c} = -\alpha\hat{i} - 2\hat{j} + \hat{k}$, where α and β are integers. If $\vec{a} \cdot \vec{b} = -1$ and $\vec{b} \cdot \vec{c} = 10$, then $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is equal to _____.

9.00

Solution:

$$\vec{a} = (1, -\alpha, \beta)$$

$$\vec{b} = (3, \beta, -\alpha)$$

$$\vec{c} = (-\alpha, -2, 1); \alpha, \beta \in I$$

$$\vec{a} \cdot \vec{b} = -1 \Rightarrow 3 - \alpha\beta - \alpha\beta = -1$$

$$\Rightarrow \alpha\beta = 2$$

$$\vec{b} \cdot \vec{c} = 10$$

$$\Rightarrow -3\alpha - 2\beta - \alpha = 10$$

$$\Rightarrow 2\alpha + \beta + 5 = 0$$

$$\therefore \alpha = -2; \beta = -1$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 2 & -2 & 1 \end{vmatrix}$$

$$= 1(-1 + 4) - 2(3 - 4) - 1(-6 + 2)$$

$$= 3 + 2 + 4 = 9$$

Q89: Let $A = \{n \in N | n^2 \leq n + 10,000\}$, $B = \{3k + 1 | k \in N\}$ and $C = \{2k | k \in N\}$, then the sum of all the elements of the set $A \cap (B - C)$ is equal to

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832.00

Solution:

$$B - C \equiv \{7, 13, 19, \dots, 97, \dots\}$$

$$\text{Now, } n^2 - n \leq 100 \times 100$$

$$\Rightarrow n(n - 1) \leq 100 \times 100$$

$$\Rightarrow A = \{1, 2, \dots, 100\}.$$

$$\text{So, } A \cap (B - C) = \{7, 13, 19, \dots, 97\}$$

$$\text{Hence, sum} = \frac{16}{2}(7 + 97) = 832.$$

Q90: Let $y = y(x)$ be the solution of the differential equation $dy = e^{\alpha x + y} dx; \alpha \in \mathbb{N}$. If $y(\log_e 2) = \log_e 2$ and $y(0) = \log_e \left(\frac{1}{2}\right)$, then the value of α is equal to _____.

Q90: Let $y = y(x)$ be the solution of the differential equation $dy = e^{\alpha x + y} dx; \alpha \in N$. If $y(\log_e 2) = \log_e 2$ and $y(0) = \log_e \left(\frac{1}{2}\right)$, then the value of α is equal to _____.

2.00

Solution:

$$\int e^{-y} dy = \int e^{\alpha x} dx$$

$$\Rightarrow -e^{-y} = \frac{e^{\alpha x}}{\alpha} + C \quad \dots (i)$$

$$\text{Put } (x, y) = (\ln 2, \ln 2)$$

$$\frac{-1}{2} = \frac{2^\alpha}{\alpha} + C \quad \dots (ii)$$

$$\text{Put } (x, y) \equiv (0, -\ln 2)$$

$$-2 = \frac{1}{\alpha} + C \quad \dots (iii)$$

$$(ii) - (iii)$$

$$\frac{2^\alpha - 1}{\alpha} = \frac{3}{2}$$

$$\Rightarrow \alpha = 2 \text{ (as } \alpha \in N)$$